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Short-distance expansions of correlation functions in the sine-Gordon theory

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Abstract

We examine the two-point correlation functions of the fields $\exp(i\alpha\Phi)$ in the sine-Gordon theory at all values of the coupling constant β . Using conformal perturbation theory, we write down explicit integral expressions for every order of the short-distance expansion. Using a novel technique analogous to dimensional regularisation, we evaluate these integrals for the first few orders finding expressions in terms of generalised hypergeometric functions. From these derived expressions, we examine the limiting forms at the points where the sine-Gordon theory maps onto a doubled Ising and the Gross-Neveu $SU(2)$ models. In this way we recover the known expansions to first subleading order of the spin and disorder fields about criticality in the Ising model and we correct an omission in past calculations of the Kosterlitz-Thouless flows in the Gross-Neveu $SU(2)$ model.

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1. Introduction

The most important generally unsolved problem in the subject of massive integrable quantum field theory in two dimensions is the computation of correlation functions. A completely general approach to the study of the large-distance expansion is well known: one can insert a multiparticle resolution of the identity between fields and thus obtain an infinite integral representation for the correlation function involving the form factors. Multiparticle form factors have been computed in a variety of models by Smirnov [1]. In principle, this form factor sum completely characterizes the correlation function. However the complexity of the form factors makes this representation difficult to utilize.

On the other hand, many quantum field theories have well-defined short-distance expansions. This short-distance expansion is essentially independent of the large-distance expansion discussed above; it is indeed extremely difficult to obtain one from the other. In this paper we develop the short-distance expansion of sine-Gordon correlation functions. This is also an infinite integral representation, though the integrals can be much more compactly written than the large-distance expansion involving the form factors. It is our aim to demonstrate that this short-distance expansion is tractable.

The form of the two-point correlation functions of the fields $\exp(i\alpha\Phi)$ in the sine-Gordon model at the free-fermion point (SG_{ff}) is well understood. Because these fields in the SG_{ff} can be mapped onto a doubling of the spin/disorder fields in the Ising model (Itzykson and Zuber [2]), the results found by Wu et al. [3] allow the expression of the correlators as solutions of a Painleve III non-linear differential equation. More recently, this result was extended in [4]. There, using a generalization of the techniques developed in [5], it was shown how correlators of the fields $\exp(i\alpha\Phi)$, $0 < \alpha < 1$, are characterized by solutions to a non-linear equation, the sinh-Gordon equation. The proof of this result lay in expressing the large-distance expansion of the correlators as Fredholm determinants through the use of form factors, and then deriving differential equations for the Fredholm determinants.

In this paper we present new results characterizing correlators of the above fields in conformal perturbation theory for values of the sine-Gordon coupling, $\hat{\beta}$, in the range $1 < \hat{\beta} < \sqrt{2}$. (Here, $\hat{\beta}$ is related to the conventional sine-Gordon coupling constant as $\hat{\beta} = \beta/\sqrt{4\pi}$, the free fermion point occurring at $\hat{\beta} = 1$.) Because of the U(1) charge symmetry of the conformal limit of the sine-Gordon theory, we focus upon the following two correlators:

$$G(\alpha, -\alpha) = \langle \exp(i\alpha\Phi(z, \bar{z})) \exp(-i\alpha\Phi(0)) \rangle, \quad (1.1)$$

$$G(\hat{\beta}/2, \hat{\beta}/2) = \langle \exp(i\hat{\beta}\Phi(z, \bar{z})/2) \exp(i\hat{\beta}\Phi(0)/2) \rangle. \quad (1.2)$$

Expanding in powers of the mass parameter, λ , we are able to write integral expressions for every order of the short-distance expansion. However, not all orders are finite for the given range of $\hat{\beta}$. For the correlator $G(\alpha, -\alpha)$, IR singularities are absent at $O(\lambda^{2n})$ given $\hat{\beta}^2 > 2 - 1/n$, while for the correlator $G(\hat{\beta}/2, \hat{\beta}/2)$, IR singularities do not appear at $O(\lambda^{2n+1})$ provided $\hat{\beta}^2 > (2n+1)/(n+1)$. UV singularities are absent at every order for both correlators provided $\hat{\beta} < \sqrt{2}$.

In principle it is possible to evaluate the integrals at every order. The singularities appear as poles in the resulting functions: at lower orders, gamma functions, at higher orders, generalised hypergeometric functions. However, the results at higher orders quickly become intractable and not particularly illuminating. Thus we restrict ourselves to evaluating the correlator, $G(\alpha, -\alpha)$, to $O(\lambda^2)$, and the correlator, $G(\hat{\beta}/2, \hat{\beta}/2)$, to $O(\lambda^3)$. Indeed, $G(\hat{\beta}/2, \hat{\beta}/2)$ at $O(\lambda^3)$ is already only finite for $\hat{\beta}^2 > 3/2$, one-half of the full range of interest, $1 < \hat{\beta}^2 < 2$.

The integral expressions in the perturbative expansion do not arise from a diagrammatic analysis. Thus it might seem necessary to explicitly subtract the contribution of the

bubble diagrams. (Without such a subtraction, the terms in the expansion are formally divergent.) However, the technique, as first introduced by Dotsenko [6], that we use to evaluate the terms renders them finite without explicit inclusion of the bubbles. This technique is akin to dimensional regularisation. Rather than analytically continuing the dimension of spacetime, we continue the parameter, $\hat{\beta}$. In this way not only are the integrals rendered finite, the bubble diagrams evaluate to zero.

At particular points in the coupling, the sine-Gordon theory is known to map onto familiar theories. At $\hat{\beta} = 1$, sine-Gordon maps onto a doubled Ising model, and at $\hat{\beta} = \sqrt{2}$, it maps onto the SU(2) Gross-Neveu model. We examine our perturbative expansion at these points. At both of these points the perturbative series become infinite and renormalisation is required. At $\hat{\beta} = 1$ we introduce a novel renormalisation scheme to remove the IR divergences. At $\hat{\beta} = \sqrt{2}$ we employ a more conventional scheme to take care of the UV divergences. However it differs from those previously used in that it involves a renormalisation of the cosine potential. Having then performed the necessary renormalisations, we obtain the known expansions to the first subleading order of correlators for the order/disorder fields in the Ising model. To lowest order we reproduce the known Kosterlitz-Thouless flows in the Gross-Neveu SU(2) model. But in carrying this calculation to one higher order, our scheme gives a different answer, a result of the cosine potential renormalisation. The Ising calculations and the lowest order SU(2) calculation provide a degree of confidence that our derived expressions for arbitrary $\hat{\beta}$ are correct.

2. Overview of the perturbative expansion

The action for the sine-Gordon theory in Euclidean space-time (our conventions are $z = (t + ix)/2$, $\bar{z} = (t - ix)/2$ and $d^2z = -dtdx/2$) is

$$S = S_{\text{CFT}} + S_{\text{Pert}} = -\frac{1}{4\pi} \int d^2z \left(\partial_z \Phi \partial_{\bar{z}} \Phi + 4\lambda : \cos(\hat{\beta}\Phi) : \right) \quad (2.1)$$

where, again, the parameter $\hat{\beta}$ is related to the conventionally normalized coupling by $\hat{\beta} = \beta/\sqrt{4\pi}$. As shown by Coleman [7], the UV divergences in the theory for values of $\hat{\beta} < \sqrt{2}$ (i.e. for values of $\hat{\beta}$ where the coupling λ is not irrelevant), may be removed by the straight-forward normal ordering of the $\cos(\hat{\beta}\Phi)$. In the process, the coupling constant λ is multiplicatively renormalised. As no wavefunction renormalisation is necessary, the scaling dimensions of the fields remain unchanged by the introduction of the cosine perturbation – the dimensions are the same as those in the deep ultraviolet. Because the structure of the field theory is the same as its conformal limit, it is meaningful to perturb about the free theory – indeed, we may use the same labels for the fields in both the free and massive theories. For a general discussion of conformal perturbation theory see [8].

The coupling constant, λ , in (2.1) can be directly related to the mass, m , of the asymptotic states, the solitons. Zamolodchikov [9], using results derived with thermo-

dynamic Bethe ansatz, demonstrated that

$$\lambda = -\tilde{\lambda}(\hat{\beta})m^{2-\hat{\beta}^2} \quad (2.2)$$

where ¹

$$\tilde{\lambda}(\hat{\beta}) = \frac{\Gamma(\hat{\beta}^2/2)}{\Gamma(1-\hat{\beta}^2/2)} \left[\frac{\sqrt{\pi}\Gamma(1/(2-\hat{\beta}^2))}{\Gamma(\hat{\beta}^2/(4-2\hat{\beta}^2))} \right]^{2-\hat{\beta}^2}. \quad (2.3)$$

We note that $\lambda = -m$ at $\hat{\beta} = 1$. This relation (2.2) will prove to be useful in our examination of the Ising model.

Consider the correlation functions

$$G(\alpha, \alpha') = \langle 0 | : \exp(i\alpha\Phi(z, \bar{z})) :: \exp(i\alpha'\Phi(0)) : | 0 \rangle. \quad (2.4)$$

These correlation functions are highly non-trivial. Even at the free-fermion point, $\hat{\beta} = 1$, the bosonisation relations, $: \exp(\pm\Phi) := \Psi_{\pm}\Psi_{\mp}$, where the Ψ 's are the spinor components in the massive Thirring model, imply the fields, $\exp(i\alpha\Phi)$, are not simply representable as fermions.

The perturbative expansion of these correlators in a path integral formulation is

$$\begin{aligned} G(\alpha, \alpha') &= \frac{\int D\Phi e^{-S_{\text{CFT}}(\Phi)} e^{\frac{\lambda}{\pi} \int d^2w \cos(\hat{\beta}\Phi(w))} e^{i\alpha\Phi(z)} e^{i\alpha'\Phi(0)}}{\int D\Phi e^{-S_{\text{CFT}}(\Phi)} e^{\frac{\lambda}{\pi} \int d^2w \cos(\hat{\beta}\Phi(w))}} \\ &= \frac{\sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{\lambda}{2\pi}\right)^n \int d^2w_i \sum_{l_k=\pm 1} \langle e^{il_1\hat{\beta}\Phi(w_1)} \dots e^{il_n\hat{\beta}\Phi(w_n)} e^{i\alpha\Phi(z)} e^{i\alpha'\Phi(0)} \rangle_{\text{CFT}}}{\sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{\lambda}{2\pi}\right)^n \int d^2w_i \sum_{l_k=\pm 1} \langle e^{il_1\hat{\beta}\Phi(w_1)} \dots e^{il_n\hat{\beta}\Phi(w_n)} \rangle_{\text{CFT}}} \end{aligned} \quad (2.5)$$

where we have expanded $e^{-S_{\text{pert}}}$ in powers of λ and $\langle \rangle_{\text{CFT}}$ indicates the correlator is to be evaluated in the conformal limit. The advantage of expressing $G(\alpha, \alpha')$ in this form is that the correlators of vertex operators in the conformal limit, i.e. $\langle \rangle_{\text{CFT}}$, are easily evaluated. The denominators of these expressions, representing the bubble diagrams in the theory, will prove to be important in proving the convergence of the terms in the perturbative expansion.

To evaluate the resulting conformal correlators we need to specify the free boson propagator. With the action as given in (2.1) the propagator is

$$\langle \Phi(z, \bar{z}) \Phi(0) \rangle = -\log(cz\bar{z}), \quad (2.6)$$

where c is some arbitrary constant. As our convention we set $c = 1$. With this propagator the OPE of two vertex operators $: e^{i\alpha\Phi(z, \bar{z})} :: e^{i\beta\Phi(0)} :$, is then equal to

$$: e^{i\alpha\Phi(z, \bar{z})} :: e^{i\beta\Phi(0)} := (|z|^2)^{\alpha\beta} : e^{i(\alpha+\beta)\Phi(z, \bar{z})} : + \dots \quad (2.7)$$

¹ Some caution must be used in applying Zamolodchikov's result directly as he uses different conventions for both the action and the propagator.

Using this OPE, the terms of the form $\langle e^{i\alpha_1\Phi(z_1)} \dots e^{i\alpha_n\Phi(z_n)} \rangle_{\text{CFT}}$ are then given by

$$\langle e^{i\alpha_1\Phi(z_1)} \dots e^{i\alpha_n\Phi(z_n)} \rangle_{\text{CFT}} = \delta\left(\sum \alpha_i\right) \prod_{i < j} (|z_i - z_j|^2)^{\alpha_i \alpha_j}. \quad (2.8)$$

That $\langle \rangle_{\text{CFT}}$ vanishes unless $\sum \alpha_i = 0$ is a consequence of the $U(1)$ symmetry of the free massless theory.

Given this $U(1)$ symmetry, only certain values of α and α' lead to non-vanishing correlators, $G(\alpha, \alpha')$, in the massive theory in perturbation theory. We examine two in particular, $G(\alpha, -\alpha)$ and $G(\hat{\beta}/2, \hat{\beta}/2)$. The first has non-vanishing terms at all even orders in the perturbation, the second at all odd orders. One reason for choosing to examine these correlators in particular is their relation to the Ising spin and disorder field correlators, $\langle \sigma(z, \bar{z}) \sigma(0) \rangle$ and $\langle \mu(z, \bar{z}) \mu(0) \rangle$, respectively. At $\hat{\beta} = 1$, these correlators are related to the G 's via

$$\langle \sigma(z, \bar{z}) \sigma(0) \rangle^2 = \langle \sin\left(\frac{\Phi(z, \bar{z})}{2}\right) \sin\left(\frac{\Phi(0)}{2}\right) \rangle = \frac{1}{2} \left(G\left(\frac{1}{2}, -\frac{1}{2}\right) - G\left(\frac{1}{2}, \frac{1}{2}\right) \right), \quad (2.9)$$

$$\langle \mu(z, \bar{z}) \mu(0) \rangle^2 = \langle \cos\left(\frac{\Phi(z, \bar{z})}{2}\right) \cos\left(\frac{\Phi(0)}{2}\right) \rangle = \frac{1}{2} \left(G\left(\frac{1}{2}, -\frac{1}{2}\right) + G\left(\frac{1}{2}, \frac{1}{2}\right) \right), \quad (2.10)$$

where we have used $G(\alpha, \alpha') = G(-\alpha, -\alpha')$.

The perturbative expansion for $G(\alpha, -\alpha)$ and $G(\hat{\beta}/2, \hat{\beta}/2)$, using (2.8) and the $U(1)$ symmetry, simplifies to

$$G(\alpha, -\alpha) = \frac{\sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{\lambda}{2\pi}\right)^{2n} \binom{2n}{n} \int d^2 w_1 \dots d^2 w_{2n} \omega(\hat{\beta}, w_i) \tau(\alpha, \hat{\beta}, w_i)}{\sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{\lambda}{2\pi}\right)^{2n} \binom{2n}{n} \int d^2 w_1 \dots d^2 w_{2n} \omega(\hat{\beta}, w_i)} \quad (2.11)$$

and

$$G\left(\frac{\hat{\beta}}{2}, \frac{\hat{\beta}}{2}\right) = \frac{\sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{\lambda}{2\pi}\right)^{2n+1} \int d^2 w_1 \dots d^2 w_{2n+1} \binom{2n+1}{n+1} \chi(\hat{\beta}, w_i) \psi(\hat{\beta}, w_i)}{\sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{\lambda}{2\pi}\right)^{2n} \binom{2n}{n} \int d^2 w_1 \dots d^2 w_{2n} \omega(\hat{\beta}, w_i)}, \quad (2.12)$$

where the functions ω , τ , χ , and ψ are given by

$$\omega(\hat{\beta}, w_i) = \prod_{\substack{1 \leq i \neq j \leq n \\ 2n \geq i \neq j > n}} (|w_i - w_j|^2)^{\hat{\beta}^2} \prod_{1 \leq i \leq n < j \leq 2n} (|w_i - w_j|^2)^{-\hat{\beta}^2}, \quad (2.13)$$

$$\tau(\alpha, \hat{\beta}, w_i) = |z|^{-2} |\alpha|^2 \prod_{1 \leq i \leq n} (|w_i - z|^2 |w_i|^{-2})^{\alpha \hat{\beta}} \prod_{n < i \leq 2n} (|w_i - z|^{-2} |w_i|^2)^{\alpha \hat{\beta}}, \quad (2.14)$$

$$\chi(\hat{\beta}, w_i) = \prod_{\substack{1 \leq i \neq j \leq n \\ 2n+1 \geq i \neq j > n}} (|w_i - w_j|^2)^{\hat{\beta}^2} \prod_{1 \leq i \leq n < j \leq 2n+1} (|w_i - w_j|^2)^{-\hat{\beta}^2}, \quad (2.15)$$

$$\psi(\hat{\beta}, w_i) = |z|^2|^{\frac{\hat{\beta}}{4}} \prod_{1 \leq i \leq n} (|w_i - z|^2 |w_i|^2)^{\frac{\hat{\beta}^2}{2}} \prod_{n < i \leq 2n+1} (|w_i - z|^2 |w_i|^2)^{-\frac{\hat{\beta}^2}{2}}. \quad (2.16)$$

3. Convergence of terms in perturbative expansion

In this section we examine the singularities that appear in the perturbative expansions of the correlators $G(\alpha, -\alpha)$ and $G(\hat{\beta}/2, \hat{\beta}/2)$. We find that $G(\alpha, -\alpha)$ at $O(\lambda^{2n})$ is UV finite for $\hat{\beta}^2 < 2$ and IR finite for $\hat{\beta}^2 > 2 - 1/n$, and that $G(\hat{\beta}/2, \hat{\beta}/2)$ at $O(\lambda^{2n+1})$ is again UV finite for $\hat{\beta}^2 < 2$ but IR finite for $\hat{\beta}^2 > (2n+1)/(n+1)$. We give a proof of this beginning with $G(\alpha, -\alpha)$.

To show UV finiteness, we consider the contribution to $G(\alpha, -\alpha)$ at $O(\lambda^{2n})$. It takes the form

$$\int d^2 w_1 \dots d^2 w_{2n} \langle e^{i\alpha\Phi(z)} e^{-i\alpha\Phi(0)} e^{i\hat{\beta}\Phi(w_1)} e^{-i\hat{\beta}\Phi(w_2)} \dots e^{i\hat{\beta}\Phi(w_{2n-1})} e^{-i\hat{\beta}\Phi(w_{2n})} \rangle$$

– disconnected pieces. (3.1)

The leading-order UV singularity occurs as $w_1 \rightarrow w_2, w_3 \rightarrow w_4, \dots, w_{2n-1} \rightarrow w_{2n}$. These singularities are governed by the operator product expansion

$$\begin{aligned} : e^{i\hat{\beta}\Phi(x)} :: e^{-i\hat{\beta}\Phi(y)} : &= |x - y|^{-2\hat{\beta}^2} : e^{i\hat{\beta}(\Phi(x) - \Phi(y))} : + \text{finite pieces} \\ &= |x - y|^{-2\hat{\beta}^2} (I - \hat{\beta}^2 |x - y|^2 : \partial_y \Phi(y) \partial_{\bar{y}} \Phi(y) : + \dots), \end{aligned} \quad (3.2)$$

where in the second line pieces such as $(x - y) \partial_y \Phi(y)$ and its antiholomorphic counterpart are discarded. Such pieces, when the OPE is inserted into the correlator, integrate out.

The leading-order singularity in this term does not come from the contribution of the identity operator in the above OPE; rather this term leads to a disconnected piece which is subsequently subtracted off. Thus the leading-order singularity that contributes is $|x - y|^{2-2\hat{\beta}^2}$. So the leading UV singularity at $O(\lambda^{2n})$ in $G(\alpha, -\alpha)$ has dimension

$$d_{UV} = n(2 - 2\hat{\beta}^2 + 2). \quad (3.3)$$

The n arises as there n approaches $w_i \rightarrow w_{i+1}$, the extra 2 from the integration $\int d^2(w_i - w_{i+1})$. Hence for $\hat{\beta}^2 < 2$, i.e. $d_{UV} > 0$, the terms are UV finite, as is expected from Coleman [7].

The reader may worry that stronger divergences are introduced by considering the approaches $w_1 \rightarrow w_3, w_5 \rightarrow w_7$, and so forth in addition to $w_1 \rightarrow w_2, w_3 \rightarrow w_4$, etc. But such additional singularities are governed by the OPE

$$\begin{aligned} : \partial_x \Phi(x) \partial_{\bar{x}} \Phi(x) :: \partial_y \Phi(y) \partial_{\bar{y}} \Phi(y) : \\ = |x - y|^{-4} (I + |x - y|^2 : \partial_y \Phi(y) \partial_{\bar{y}} \Phi(y) : + \dots). \end{aligned} \quad (3.4)$$

Again the leading term in the OPE is disconnected. So its leading singular contribution is $|x - y|^{-2}$. But this is exactly canceled by the integration $\int d^2(x - y)$ and so the overall UV singularity becomes no greater.

When all the w_i 's are allowed to approach one another, successively substituting the above OPEs into the correlator in (3.1) leads one eventually to the correlator

$$\langle e^{i\alpha\Phi(z)} e^{-\alpha\Phi(0)} : \partial_w \Phi(w) \partial_{\bar{w}} \Phi(\bar{w}) : \rangle. \quad (3.5)$$

But this correlator, as can be easily checked, introduces no additional UV singularities as w approaches z or 0. Thus allowing in addition the w_i 's to approach z or 0 does not change the UV properties of the perturbative expansion.

To demonstrate that $G(\alpha, -\alpha)$ is IR finite at $O(\lambda^{2n})$ for $\hat{\beta}^2 > 2 - 1/n$, we recast each term in the perturbative expansion by making the conformal transformation

$$w_i \rightarrow 1/w_i. \quad (3.6)$$

The perturbative terms then appear as

$$\begin{aligned} & \int |w_1|^{2\hat{\beta}^2-4} d^2 w_1 \dots |w_{2n}|^{2\hat{\beta}^2-4} d^2 w_{2n} \\ & \times \langle e^{i\alpha\Phi(z)} e^{-i\alpha\Phi(0)} e^{i\hat{\beta}\Phi(w_1)} e^{-i\hat{\beta}\Phi(w_2)} \dots e^{i\hat{\beta}\Phi(w_{2n-1})} e^{-i\hat{\beta}\Phi(w_{2n})} \rangle \\ & - \text{disconnected pieces.} \end{aligned} \quad (3.7)$$

The IR behaviour of this integral is determined by sending $w_i \rightarrow 0$ for all i . The singular behaviour of the correlator in the above integral can be extracted by first sending $w_1 \rightarrow w_2$, $w_3 \rightarrow w_4$, $\dots$, $w_{2n-1} \rightarrow w_{2n}$, then $w_1 \rightarrow w_3$, $w_5 \rightarrow w_7$, $\dots$, $w_{2n-3} \rightarrow w_{2n-1}$, and so on. Using the OPEs given in (3.2) and (3.4) and discarding, as before, the disconnected pieces associated with the identity, the dimension of the leading IR singularity is

$$d_{\text{IR}} = [2n(2\hat{\beta}^2 - 2)] + [2 - 2n\hat{\beta}^2]. \quad (3.8)$$

The latter term in d_{IR} is the contribution coming from the correlator, the former term from $\int |w_i|^{2\hat{\beta}^2-4} d^2 w_i$. It is important to not include the singularities that arise from the OPEs involving $e^{i\alpha\Phi(0)}$. These singularities are properly UV. Then for the term to be IR finite, we need $d_{\text{IR}} > 0$. Thus $\hat{\beta}^2$ must be greater than $2 - 1/n$.

To check the UV finiteness of $G(\hat{\beta}/2, \hat{\beta}/2)$, a similar argument is used to the one above. At $O(\lambda^{2n+1})$ the term in the perturbative expansion appears as

$$\begin{aligned} & \int d^2 w_1 \dots d^2 w_{2n+1} \\ & \times \langle e^{i\hat{\beta}/2\Phi(z)} e^{i\hat{\beta}/2\Phi(0)} e^{-i\hat{\beta}\Phi(w_1)} e^{i\hat{\beta}\Phi(w_2)} e^{-i\hat{\beta}\Phi(w_3)} \dots e^{i\hat{\beta}\Phi(w_{2n})} e^{-i\hat{\beta}\Phi(w_{2n+1})} \rangle \\ & - \text{disconnected pieces.} \end{aligned} \quad (3.9)$$

The most singular pieces of this expression come when $w_2 \rightarrow w_3$, $w_4 \rightarrow w_5$, $\dots$, $w_{2n} \rightarrow w_{2n+1}$. These approaches are governed by the OPE in (3.2). Thus the overall dimension of the leading UV singularity is

$$d_{UV} = n(4 - 2\hat{\beta}^2), \quad (3.10)$$

and so again for $\hat{\beta}^2 < 2$ the terms are UV finite. Additional approaches, say $w_2 \rightarrow w_4$, $w_6 \rightarrow w_8$, etc., are governed by the OPE (3.4) and so, as before, do not change d_{UV} . If all the w_i 's, $i > 1$, are allowed to approach one another, say at w , the succession of OPEs leaves one to evaluate the correlator

$$\langle e^{i\hat{\beta}/2\Phi(z)} e^{i\hat{\beta}/2\Phi(0)} e^{-i\beta\Phi(w_1)} : \partial_w \Phi(w) \partial_{\bar{w}} \Phi(w) : \rangle. \quad (3.11)$$

But as $w \rightarrow w_1$, 0, or z , no new UV singularities appear.

To obtain the IR behaviour of $G(\hat{\beta}/2, \hat{\beta}/2)$ for $n > 0$, we begin by treating the term at $O(\lambda^{2n+1})$ like the term at $O(\lambda^{2n})$ in $G(\alpha, -\alpha)$, contracting the last $2n$ vertex operators $e^{\pm i\hat{\beta}\Phi(w)}$ in the correlator. However there is an additional contribution of $2\hat{\beta}^2 - 4$ to d_{IR} in this case that comes from letting $w \rightarrow w_1$ in (3.11). Thus the term is IR finite if $\hat{\beta}^2 > (2n+1)/(n+1)$. For $n = 0$ it is easy to see that this formula still holds: the term at $O(\lambda)$ is IR finite if $\hat{\beta}^2 > 1$.

Though we have identified the ranges of $\hat{\beta}$ where the perturbative expansion is divergent, we point out that the expressions we derive in the next section will only be divergent for individual values of $\hat{\beta}$: in general the expressions we will derive will be meromorphic functions of $\hat{\beta}$. However it is not clear how to interpret the analytically continued expressions in regions of $\hat{\beta}$ where the original integral expressions are divergent. It is possible that this continuation does not capture all the relevant physics. For example at $\hat{\beta}^2 < 1$, it may be the continued correlators do not reflect the presence of quantum breathers.

A note of caution on the use of the results of this section: they apply only to connected correlators. (In the above analysis only disconnected bubble diagrams have been subtracted.) Disconnected divergences still lurk in the results that follow, generally appearing where $\alpha\hat{\beta}$ in (3.1) is an integer. We will find ourselves dealing with them at $\hat{\beta}^2 = 2$ in our evaluation of the Gross-Neveu model.

4. Evaluation of terms of perturbative series

We begin by focusing on the integrals in the numerator of the perturbative expansion (i.e. we ignore the contribution of the bubble diagrams). Although these integrals are formally divergent, the techniques (as first developed by Dotsenko [6]) we employ to evaluate them render them finite in a manner somewhat analogous to dimensional regularisation. This handling of the infinities suggests the technique takes into account the bubble diagrams. This is born out by the fact that when the contribution of the bubbles are calculated explicitly using the same techniques, they turn out to be identically zero.

The integrals we thus consider are of the form

$$I_{2n} = \int d^2 w_1 \cdots d^2 w_{2n} \omega(\hat{\beta}, w_1, \dots, w_{2n}) \tau(\alpha, \hat{\beta}, w_1, \dots, w_{2n}), \quad (4.1)$$

$$I_{2n+1} = \int d^2 w_1 \cdots d^2 w_{2n+1} \chi(\hat{\beta}, w_1, \dots, w_{2n+1}) \psi(\hat{\beta}, w_1, \dots, w_{2n+1}). \quad (4.2)$$

These integrals are, in their full generality, unwieldy. Thus, in the end, we only explicitly evaluate I_1 , I_2 , and I_3 . Indeed, I_3 is already only finite for $\hat{\beta}^2 > 3/2$, a part of the full range of interest. We begin with I_1 .

4.1. Evaluation of I_1

To make sense of I_1 we must first abandon light cone coordinates, writing it in the usual x, t coordinates. Doing so, and at the same time scaling out the z -dependence, we obtain

$$\begin{aligned} I_1 &= \int d^2 w_1 |z|^{\hat{\beta}^2/2} |w_1 - z|^{-\hat{\beta}^2} |w_1|^{-\hat{\beta}^2} \\ &= -2|z|^{2-3\hat{\beta}^2/2} \int dt_1 dx_1 ((t_1 - 1)^2 + x_1^2)^{-\hat{\beta}^2/2} (t_1^2 + x_1^2)^{-\hat{\beta}^2/2}. \end{aligned} \quad (4.3)$$

Now the branch points in I_1 's integrand lie on the imaginary axis of x_1 . Thus the deformation of the x_1 -contour, $x_1 \rightarrow -ie^{-i2\epsilon}$, $\epsilon \ll 1$ is permissible. After this deformation, the terms in the integral can be factorized through the change of variables,

$$w_1^\pm = t_1 \pm x_1.$$

I_1 is then left in the form

$$I_1 = -i|z|^{2-3\hat{\beta}^2/2} \int dw_1^+ dw_1^- \theta(w_1^+, \epsilon) \theta(w_1^-, -\epsilon), \quad (4.4)$$

where $\theta(w, \epsilon) = (w - 1 - i\epsilon\Delta)^{\hat{\beta}^2/2} (w - i\epsilon\Delta)^{\hat{\beta}^2/2}$ and $\Delta = w^+ - w^-$.

Having made these transformations, we are faced with an integral represented by a contour deformed around the branch points of algebraic functions. The signs of the coefficients of ϵ in the integrand tell us how to make the deformations. Focusing on the w_1^- -contour, the coefficients of the relevant ϵ 's are functions of the variables w_1^+ . For various values of w_1^+ 's, the w_1^- -contour will enclose no poles, and the corresponding contribution to I_1 is 0. It is easy enough to see a contribution is only made when $0 < w_1^+ < 1$.

When w_1^+ does fall in this range, the w_1^- -contour appears as drawn at the top of Fig. 1. Deforming this contour about the branch point at 1 introduces the phases as given in the bottom of Fig. 1. I_1 is then simply

$$\begin{aligned} I_1 &= -2|z|^{2-3\hat{\beta}^2/2} s(\hat{\beta}^2/2) \int_0^1 dw_1 w_1^{-\hat{\beta}^2/2} (1 - w_1)^{-\hat{\beta}^2/2} \int_1^\infty dw_1 w_1^{-\hat{\beta}^2/2} (w_1 - 1)^{-\hat{\beta}^2/2} \\ &= -2|z|^{2-3\hat{\beta}^2/2} s(\hat{\beta}^2/2) B(1 - \tfrac{1}{2}\hat{\beta}^2, 1 - \tfrac{1}{2}\hat{\beta}^2) B(\hat{\beta}^2 - 1, 1 - \tfrac{1}{2}\hat{\beta}^2), \end{aligned} \quad (4.5)$$

where we have introduced the notation $s(a) \equiv \sin(\pi a)$ and in the last line employed the standard beta-function representations of these integrals. In writing this expression for I_1 ,

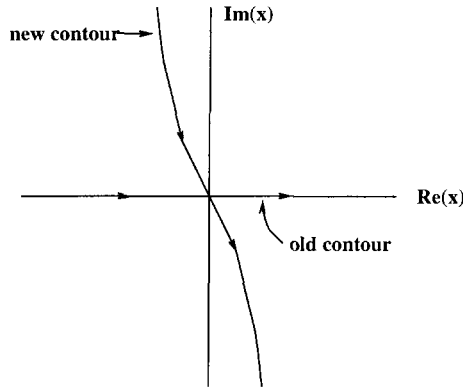


Fig. 1. Shown is the contour used in the evaluation of I_1 .

we have assumed that the portions of the contours that circle the branch point contribute nothing. This assumption can always be made good by continuing the exponents of the terms in the integrands to points where these portions do contribute nothing, evaluating the integral, and then continuing back to the desired value of the exponents in the final expression. It is this continuation of the exponents that makes the method akin to dimensional regularisation. But instead of continuing the dimension of space-time, we continue the parameter $\hat{\beta}$, i.e. the scaling dimension of the vertex operators appearing in the action.

We mention that the integral I_1 is one that arises in calculating the four-point Virasoro-Shapiro formula. The methods used in this case for evaluating this integral give the same answer as above.

4.2. Evaluation of I_2

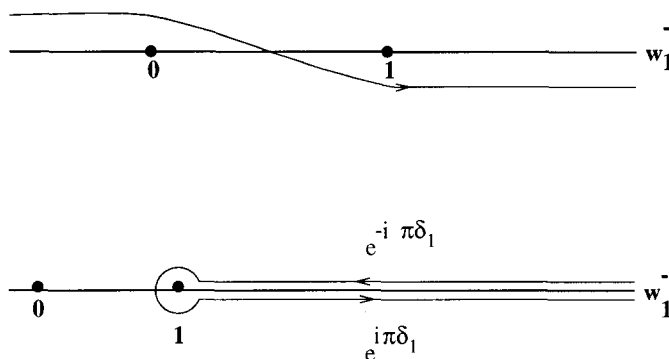
We make the same set of transformations on I_2 that allowed us to write I_1 as in (4.4):

$$I_2 = -|z|^{4-2\alpha^2-2\hat{\beta}^2} \int dw_1^+ dw_2^+ \theta(w_1^+, w_2^+, \epsilon) \int dw_1^- dw_2^- \theta(w_1^-, w_2^-, -\epsilon), \quad (4.6)$$

where

$$\begin{aligned} \theta(w_1, w_2, \epsilon) = & (w_1 - 1 - i\epsilon\Delta_1)^{\alpha\hat{\beta}} (w_1 - i\epsilon\Delta_1)^{-\alpha\hat{\beta}} (w_1 - w_2 - i\epsilon(\Delta_1 - \Delta_2))^{-\hat{\beta}^2} \\ & \times (w_2 - 1 - i\epsilon\Delta_2)^{-\alpha\hat{\beta}} (w_2 - i\epsilon\Delta_2)^{\alpha\hat{\beta}}. \end{aligned} \quad (4.7)$$

In the same fashion as before, a contribution is made to I_2 only when $w_1, w_2 \in [0, 1]$. For this range, the two w^- -contours appear as in the top of Fig. 2. In this figure we have assumed that $w_{i_1}^+ > w_{i_2}^+$ and assumed the $w_{i_1}^-$ -integral is done last. Deforming the contours about the branch points at 1 gives the picture at the bottom of Fig. 2. Because the $w_{i_2}^-$ integrand has a cut at $w_{i_1}^-$, the phase assignment associated with the deformation is more complicated than with I_1 .

Fig. 2. Shown are the contours used in the evaluation of I_2 .

In addition to the branch points that arise directly from the terms of the form $(w_i - a + i\epsilon f(w_k^+, w_k^-))$, branch points also appear because of prior integrations. After performing the w_{i1}^- integration, I_2 takes the form

$$I_2 = \int dw_1^+ dw_2^+ \theta(w_1^+, w_2^+, \epsilon) \times \int dw_2^- (w_2^- - 1 + i\epsilon\Delta_{i2})^{-\alpha\hat{\beta}} (w_2^- + i\epsilon\Delta_{i2})^{\alpha\hat{\beta}} g(w_{i2}^- - i\epsilon), \quad (4.8)$$

where g is a hypergeometric function with branch points at 0 and 1. These branch points arise from integrating the term $(w_{i1}^- - w_{i2}^- + i\epsilon(\Delta_{i1} - \Delta_{i2}))^\gamma$. As the coefficient of ϵ in this term is positive when $w_{i1}^- \equiv w_{i2}^-$ (as $w_{i1}^+ - w_{i2}^+$), g depends on $w_{i2}^- - i\epsilon$ as indicated. The importance of all of this is, of course, that g 's branch points do not interfere with the deformation done to obtain the bottom part of Fig. 2. Nor do these branch points interfere with the deformation of the contours when w_1^+ or w_2^+ are not in the range $[0, 1]$: these branch points notwithstanding, the contribution is still zero.

The integral I_2 then takes the form

$$I_2 = -|z|^{4-2\alpha^2-2\hat{\beta}^2} [J_1(1, 2)J_2(1, 2) + 1 \leftrightarrow 2]. \quad (4.9)$$

$J_1(i_1, i_2)$ is given by

$$\begin{aligned} J_1(i_1, i_2) &= \int_0^1 dw_{i1} w_{i1}^{-\gamma_{i1}} (1 - w_{i1})^{\gamma_{i1}} \int_0^{w_{i1}} dw_{i2} w_{i2}^{-\gamma_{i2}} (1 - w_{i2})^{\gamma_{i2}} (w_{i1} - w_{i2})^{-\hat{\beta}^2} \\ &= B(1 + \gamma_{i1}, 2 - \hat{\beta}^2) B(1 - \hat{\beta}^2, 1 - \gamma_{i2}) \\ &\quad \times {}_3F_2(-\gamma_{i2}, 2 - \hat{\beta}^2, 1 - \gamma_{i2}, 3 - \hat{\beta}^2 + \gamma_{i1}, 2 - \hat{\beta}^2 - \gamma_{i2}, 1), \end{aligned} \quad (4.10)$$

where we have defined $\gamma_1 = -\gamma_2 = \alpha\hat{\beta}$. In the last line we have used the expansion

$$(1 - w)^a = \sum_{k=0}^{\infty} \frac{(-a)_k}{k!} w^k, \quad (4.11)$$

where $(x)_k \equiv x(x+1) \cdots (x+k-1) \equiv \Gamma(x+k)/\Gamma(x)$ to evaluate the integral. In this expression ${}_3F_2$ is a standard hypergeometric function. $J_2(i_1, i_2)$ takes the form

$$J_2(i_1, i_2) = 2is(\gamma_{i_1}) \int_1^\infty dw_{i_1} (w_{i_1} - 1)^{\gamma_{i_1}} w_{i_1}^{-\gamma_{i_1}} \\ \times \left[2is(\gamma_{i_2}) \int_1^{w_{i_1}} dw_{i_2} (w_{i_2} - 1)^{\gamma_{i_2}} w_{i_2}^{-\gamma_{i_2}} (w_{i_1} - w_{i_2})^{-\hat{\beta}^2} \right. \\ \left. + 2is(\gamma_{i_2} - \hat{\beta}^2) \int_{w_{i_1}}^\infty dw_{i_2} (w_{i_2} - 1)^{\gamma_{i_2}} w_{i_2}^{-\gamma_{i_2}} (w_{i_1} - w_{i_2})^{-\hat{\beta}^2} \right]. \quad (4.12)$$

Making the change of variables, $w_i \rightarrow 1/w_i$, the above may be written as

$$J_2(i_1, i_2) = -4 \left[s(\gamma_{i_1})s(\gamma_{i_2})K(i_2, i_1) + s(\gamma_{i_1})s(\gamma_{i_2} + \gamma_{i_1 i_2})K(i_1, i_2) \right], \quad (4.13)$$

where $K(i, j)$ is defined to be

$$K(i, j) = \int_0^1 dw_i w_i^{-2+\hat{\beta}^2} (1-w_i)^{\gamma_i} \int_0^{w_i} dw_j w_j^{-2+\hat{\beta}^2} (1-w_j)^{\gamma_j} (w_i - w_j)^{-\hat{\beta}^2}, \quad (4.14)$$

where we have used the fact that $\gamma_{i_2} = \hat{\beta}^2$ and that $\gamma_{i_1} = -\gamma_{i_2}$. We evaluate $K(i, j)$ in Appendix A with the result

$$K(i_1, i_2) = B(1 + \gamma_{i_1}, \hat{\beta}^2 - 1) \Gamma(1 - \hat{\beta}^2) \Gamma(\hat{\beta}^2 - 1) (-\gamma_{i_2}) (\hat{\beta}^2 - 1) \\ \times {}_3F_2(1 - \gamma_{i_2}, \hat{\beta}^2 - 1, \hat{\beta}^2, \hat{\beta}^2 + \gamma_{i_1}, 2, 1). \quad (4.15)$$

So I_2 has the final form

$$I_2 = -4|z|^{4-2\alpha^2-2\hat{\beta}^2} \left[J_1(1, 2) \left(s^2(\alpha\hat{\beta})K(2, 1) + s(\alpha\hat{\beta})s(\hat{\beta}^2 + \alpha\hat{\beta})K(1, 2) \right) \right. \\ \left. + J_1(2, 1) \left(s^2(\alpha\hat{\beta})K(1, 2) + s(\alpha\hat{\beta})s(\alpha\hat{\beta} - \hat{\beta}^2)K(2, 1) \right) \right]. \quad (4.16)$$

4.3. Evaluation of I_3

I_3 may be written in a form akin to (4.9) for I_2 :

$$I_3 = i|z|^{6-5\hat{\beta}^2/2} \sum_{\sigma \in S_3} J_1(\sigma(1), \sigma(2), \sigma(3)) J_2(\sigma(1), \sigma(2), \sigma(3)). \quad (4.17)$$

Defining $\delta_1 = \delta_2 = -\delta_3 = -\hat{\beta}^2/2$, $\delta_{12} = -\delta_{13} = -\delta_{23} = \hat{\beta}^2$, and $\delta_{ij} = \delta_{ji}$, J_1 is given by

$$J_1(i_1, i_2, i_3) = \int_0^1 dw_{i_1} w_{i_1}^{\delta_{i_1}} (1 - w_{i_1})^{\delta_{i_1}} \int_0^{w_{i_1}} dw_{i_2} w_{i_2}^{\delta_{i_2}} (1 - w_{i_2})^{\delta_{i_2}} (w_{i_1} - w_{i_2})^{\delta_{i_1/2}} \\ \times \int_0^{w_{i_2}} dw_{i_3} w_{i_3}^{\delta_{i_3}} (1 - w_{i_3})^{\delta_{i_3}} (w_{i_1} - w_{i_3})^{\delta_{i_1/3}} (w_{i_2} - w_{i_3})^{\delta_{i_2/3}}. \quad (4.18)$$

J_2 is represented by the contours in Fig. 3. The phases associated with these integrals is considerably more complicated than previous as the phases depend on the ordering of w_{i_1} and w_{i_2} . Defining $K(i_1, i_2, i_3)$ by

$$K(i_1, i_2, i_3) = \int_1^\infty dw_{i_1} w_{i_1}^{\delta_{i_1}} (w_{i_1} - 1)^{\delta_{i_1}} \int_1^{w_{i_1}} dw_{i_2} w_{i_2}^{\delta_{i_2}} (w_{i_2} - 1)^{\delta_{i_2}} (w_{i_1} - w_{i_2})^{\delta_{i_1} \delta_{i_2}} \\ \times \int_1^{w_{i_2}} dw_{i_3} w_{i_3}^{\delta_{i_3}} (w_{i_3} - 1)^{\delta_{i_3}} (w_{i_1} - w_{i_3})^{\delta_{i_1/3}} (w_{i_2} - w_{i_3})^{\delta_{i_2/3}}, \quad (4.19)$$

which after a change of variables $w_i \rightarrow 1/w_i$ can be written as

$$K(i_1, i_2, i_3) = \int_0^1 dw_{i_3} w_{i_3}^{-2-2\delta_{i_3}-\delta_{i_3 i_2}-\delta_{i_3 i_1}} (1 - w_{i_3})^{\delta_{i_3}} \\ \times \int_0^{w_{i_3}} dw_{i_2} w_{i_2}^{-2-2\delta_{i_2}-\delta_{i_2/3}-\delta_{i_2 i_1}} (1 - w_{i_2})^{\delta_{i_2}} (w_{i_3} - w_{i_2})^{\delta_{i_3 i_2}} \\ \times \int_0^{w_{i_2}} dw_{i_1} w_{i_1}^{-2-2\delta_{i_1}-\delta_{i_1 i_3}-\delta_{i_1 i_2}} (1 - w_{i_1})^{\delta_{i_1}} (w_{i_3} - w_{i_1})^{\delta_{i_3 i_1}} (w_{i_2} - w_{i_1})^{\delta_{i_1 i_2}}, \quad (4.20)$$

allows us to write $J_2(i_1, i_2, i_3)$ as

$$J_2(i_1, i_2, i_3) = (2i)^3 \left[s(\delta_{i_1}) s(\delta_{i_2}) s(\delta_{i_3}) K(i_1, i_2, i_3) \right. \\ + s(\delta_{i_1}) s(\delta_{i_2}) s(\delta_{i_3} + \delta_{i_3 i_2}) K(i_1, i_3, i_2) \\ + s(\delta_{i_1}) s(\delta_{i_2}) s(\delta_{i_3} + \delta_{i_3 i_2} + \delta_{i_3 i_1}) K(i_3, i_1, i_2) \\ + s(\delta_{i_1}) s(\delta_{i_2} + \delta_{i_1 i_2}) s(\delta_{i_3}) K(i_2, i_1, i_3) \\ + s(\delta_{i_1}) s(\delta_{i_2} + \delta_{i_1 i_2}) s(\delta_{i_3} + \delta_{i_3 i_1}) K(i_2, i_3, i_1) \\ \left. + s(\delta_{i_1}) s(\delta_{i_2} + \delta_{i_1 i_2}) s(\delta_{i_3} + \delta_{i_3 i_1} + \delta_{i_3 i_2}) K(i_3, i_2, i_1) \right]. \quad (4.21)$$

Using the fact that both J_1 and J_2 are invariant under permutations of the i 's involving 1 and 2, I_3 can be shown to take the form:

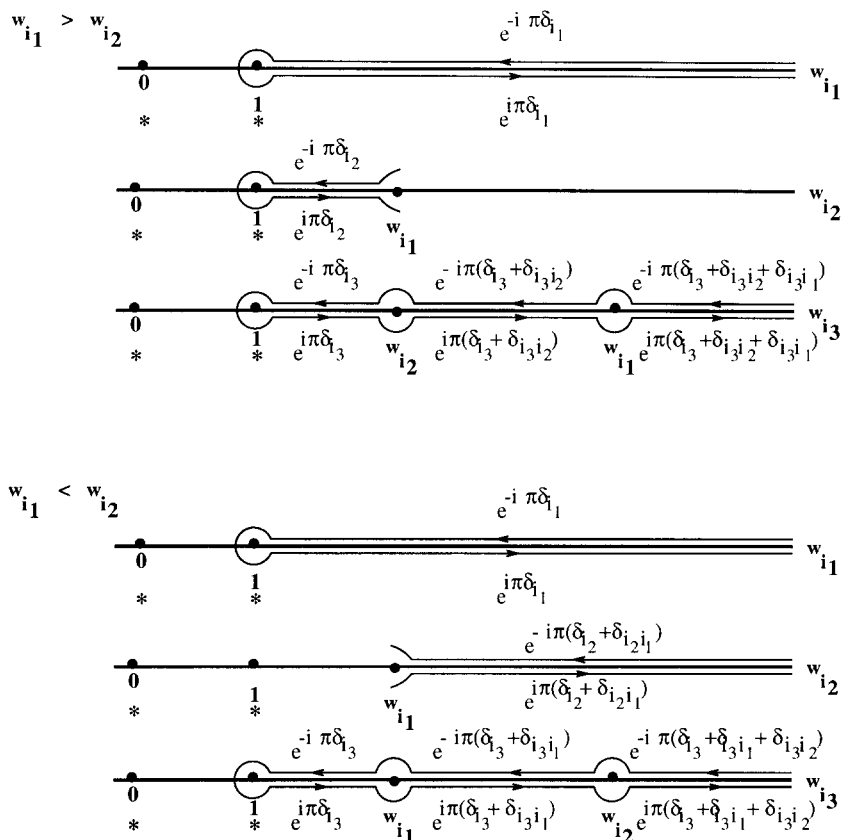


Fig. 3. The integrals in each of the cases ($w_{i1} > w_{i2}$ and $w_{i2} > w_{i1}$) are to be done in ascending order. $J_2(i_1, i_2, i_3)$ is then the sum of both cases.

$$\begin{aligned}
 I_3 = 16|z|^{6-7\hat{\beta}^2/2} & \left[K(1, 2, 3) (s^2(\hat{\beta}^2/2)(s(3\hat{\beta}^2/2) + s(\hat{\beta}^2/2))J_1(1, 3, 2) \right. \\
 & + s(\hat{\beta}^2/2)s(3\hat{\beta}^2/2)(s(3\hat{\beta}^2/2) + s(\hat{\beta}^2/2))J_1(3, 1, 2)) \\
 & \left. + K(1, 3, 2) (s^2(\hat{\beta}^2/2)(s(3\hat{\beta}^2/2) + s(\hat{\beta}^2/2))J_1(3, 1, 2)) \right]. \quad (4.22)
 \end{aligned}$$

Now all that remains is to evaluate the J_1 's and the K 's.

Using the expansion in (4.11) we find

$$\begin{aligned}
 K(1, 2, 3) &= B(1 + \hat{\beta}^2/2, -3 + 2\hat{\beta}^2)B(1 - \hat{\beta}^2, -2 + 3\hat{\beta}^2)B(1 + \hat{\beta}^2, \hat{\beta}^2 - 1) \\
 &\times \sum_{k_1 k_2 k_3}^{\infty} \frac{(\hat{\beta}^2/2)_{k_1}(\hat{\beta}^2/2)_{k_2}(\hat{\beta}^2)_{k_3}}{k_1!k_2!k_3!} \frac{(-3 + 2\hat{\beta}^2)_{k_1+k_2}}{(-2 + 5\hat{\beta}^2/2)_{k_1+k_2}} \\
 &\times \frac{(-2 + 3\hat{\beta}^2)_{k_1+k_2+k_3}}{(-1 + 2\hat{\beta}^2)_{k_1+k_2+k_3}} \frac{(-1 + \hat{\beta}^2)_{k_2+k_3}}{(2\hat{\beta}^2)_{k_2+k_3}}; \quad (4.23)
 \end{aligned}$$

$$\begin{aligned}
K(1, 3, 2) = & B(1 - \hat{\beta}^2/2, -3 + 2\hat{\beta}^2) \Gamma(1 - \hat{\beta}^2) \Gamma(-2 + \hat{\beta}^2) \Gamma(1 - \hat{\beta}^2) \Gamma(\hat{\beta}^2 - 1) \\
& \times \sum_{k_1 k_2 k_3}^{\infty} \frac{(-\hat{\beta}^2/2)_{k_1} (\hat{\beta}^2/2)_{k_2} (-\hat{\beta}^2)_{k_3}}{k_1! k_2! k_3!} \frac{(-3 + 2\hat{\beta}^2)_{k_1+k_2}}{(-2 + 3\hat{\beta}^2/2)_{k_1+k_2}} \\
& \times \frac{(-2 + \hat{\beta}^2)_{k_1+k_2+k_3}}{\Gamma(-1 + k_1 + k_2 + k_3)} \frac{(-1 + \hat{\beta}^2)_{k_2+k_3}}{\Gamma(k_2 + k_3)}; \quad (4.24)
\end{aligned}$$

$$\begin{aligned}
J_1(1, 3, 2) = & B(1 - \hat{\beta}^2/2, 3 - 3\hat{\beta}^2/2) B(1 - \hat{\beta}^2, 2 - \hat{\beta}^2) B(1 - \hat{\beta}^2, 1 - \hat{\beta}^2/2) \\
& \times \sum_{k_1 k_2 k_3}^{\infty} \frac{(-\hat{\beta}^2/2)_{k_1} (\hat{\beta}^2/2)_{k_2} (-\hat{\beta}^2)_{k_3}}{k_1! k_2! k_3!} \frac{(3 - 3\hat{\beta}^2/2)_{k_1+k_2}}{(4 - 2\hat{\beta}^2)_{k_1+k_2}} \\
& \times \frac{(2 - \hat{\beta}^2)_{k_1+k_2+k_3}}{(3 - 2\hat{\beta}^2)_{k_1+k_2+k_3}} \frac{(1 - \hat{\beta}^2/2)_{k_2+k_3}}{(2 - 3\hat{\beta}^2/2)_{k_2+k_3}}; \quad (4.25)
\end{aligned}$$

$$\begin{aligned}
J_1(3, 1, 2) = & B(1 + \hat{\beta}^2/2, 3 - 3\hat{\beta}^2/2) B(1 - \hat{\beta}^2, 2) B(1 + \hat{\beta}^2, 1 - \hat{\beta}^2/2) \\
& \times \sum_{k_1 k_2 k_3}^{\infty} \frac{(\hat{\beta}^2/2)_{k_1} (\hat{\beta}^2/2)_{k_2} (\hat{\beta}^2)_{k_3}}{k_1! k_2! k_3!} \frac{(3 - 3\hat{\beta}^2/2)_{k_1+k_2}}{(4 - \hat{\beta}^2)_{k_1+k_2}} \\
& \times \frac{(2)_{k_1+k_2+k_3}}{(3 - \hat{\beta}^2)_{k_1+k_2+k_3}} \frac{(1 - \hat{\beta}^2/2)_{k_2+k_3}}{(2 + \hat{\beta}^2/2)_{k_2+k_3}}. \quad (4.26)
\end{aligned}$$

In evaluating these integrals the same sort of techniques that went into evaluating $K(i_1, i_2)$ (i.e. handling the appearance of a pole multiplying a zero – see Appendix A) were employed. This completes the evaluation of I_3 .

4.4. Evaluation of the bubbles

We now examine the contribution of the bubble diagrams (i.e. the integrals in the denominator of the perturbative expansion). We will show that with our techniques for evaluating the integrals, the bubbles vanish identically. The integrals that we must evaluate are

$$I_{2n}^B = \int d^2 w_1 \cdots d^2 w_{2n} \prod_{\substack{1 \leq i \neq j \leq n \\ 2n \geq i \neq j > n}} (|w_i - w_j|^2)^{\hat{\beta}^2} \prod_{1 \leq i \leq n < j \leq 2n} (|w_i - w_j|^2)^{-\hat{\beta}^2}. \quad (4.27)$$

Performing the change of variables $x_i \rightarrow -ie^{-i2\epsilon} x_i$, $w_i^{\pm} = t_i \pm x_i$ as before, I_{2n}^B becomes

$$I_{2n}^B = (-i)^{2n} \int dw_i^+ dw_i^- \eta(w_1^+, \dots, w_{2n}^+, \epsilon) \eta(w_1^-, \dots, w_{2n}^-, -\epsilon),$$

where η takes the form

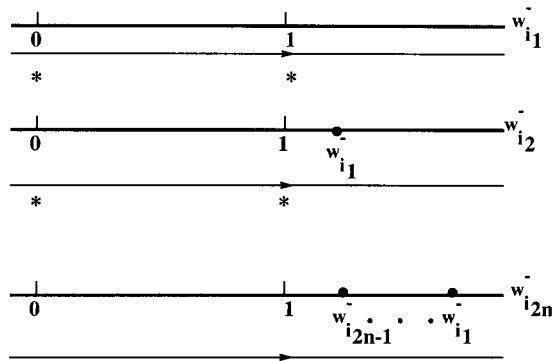


Fig. 4. Pictured are the set of contours for w_{i1}^- arising from the integration of the integral I_{2n}^B . In contrast to the situation for I_{2n} and I_{2n+1} , there are no branch points at 0 and 1.

$$\eta(w_1, \dots, w_{2n}, \epsilon) = \prod_{i=1}^{2n} \prod_{j=i+1}^n (w_i - w_j - i\epsilon(\Delta_i - \Delta_j))^{\pm \hat{\beta}^2}, \quad (4.28)$$

where the sign of the exponents depends upon the particular values of i and j . Taking the ordering of the w_i^+ 's to be $w_{i1}^+ > \dots > w_{i2n}^+$, the deformation of the w_i^- contours about the branch points then appear as in Fig. 4.

Because there are no branch points at 0 and 1, the w_{i1}^- -contour (or, indeed, the w_{i2n}^- -contour) can always be closed at ∞ . Thus I_{2n}^B is identically 0. In Fig. 4 a specific ordering of the w_i^- branch points has been assumed. However, changing this ordering does not change the argument just made. The vanishing of the bubbles is supported by our analysis of the Ising spin correlators. Because our results match previous calculations, it is likely that we have accurately taken into account the perturbative structure of the vacuum.

4.5. Summary of results

At this point we gather previous results to provide expressions for $G(\alpha, -\alpha)$ and $G(\hat{\beta}/2, \hat{\beta}/2)$:

$$\begin{aligned} G(\alpha, -\alpha) &= |z|^{-2\alpha^2} + \left(\frac{\lambda}{2\pi}\right)^2 I_2 \\ &= |z|^{-2\alpha^2} - \left(\frac{\lambda}{\pi}\right)^2 |z|^{4-2\alpha^2-2\hat{\beta}^2} \\ &\quad \times \left[J_1(1, 2) \left(s^2(\alpha\hat{\beta}) K(2, 1) + s(\alpha\hat{\beta}) s(\hat{\beta}^2 + \alpha\hat{\beta}) K(1, 2) \right) \right. \\ &\quad \left. + J_1(2, 1) \left(s^2(\alpha\hat{\beta}) K(1, 2) + s(\alpha\hat{\beta}) s(\alpha\hat{\beta} - \hat{\beta}^2) K(2, 1) \right) \right]; \end{aligned} \quad (4.29)$$

$$\begin{aligned}
G\left(\frac{\hat{\beta}}{2}, \frac{\hat{\beta}}{2}\right) &= \left(\frac{\lambda}{2\pi}\right) I_1 + \frac{1}{2} \left(\frac{\lambda}{2\pi}\right)^3 I_3 \\
&= -\frac{\lambda}{\pi} |z|^{2-3\hat{\beta}^2/2} s(\hat{\beta}^2/2) \mathbf{B}(1 - \hat{\beta}^2/2, 1 - \hat{\beta}^2/2) \mathbf{B}(\hat{\beta}^2 - 1, 1 - \hat{\beta}^2/2) \\
&\quad + \left(\frac{\lambda}{\pi}\right)^3 |z|^{6-7\hat{\beta}^2/2} \\
&\quad \times \left[K(1, 2, 3) \left(s^2(\hat{\beta}^2/2) (s(3\hat{\beta}^2/2) + s(\hat{\beta}^2/2)) J_1(1, 3, 2) \right. \right. \\
&\quad \left. \left. + s(\hat{\beta}^2/2) s(3\hat{\beta}^2/2) (s(3\hat{\beta}^2/2) + s(\hat{\beta}^2/2)) J_1(3, 1, 2) \right) \right. \\
&\quad \left. + K(1, 3, 2) \left(s^2(\hat{\beta}^2/2) (s(3\hat{\beta}^2/2) + s(\hat{\beta}^2/2)) J_1(3, 1, 2) \right) \right].
\end{aligned} \tag{4.30}$$

The expressions for $J(i, j)$ and $K(i, j)$ are given in (4.10) and (4.15) respectively. The expressions for $J(i, j, k)$ and $K(i, j, k)$ are given in (4.23) through (4.26).

5. Sine-Gordon as Ising and SU(2) Gross-Neveu

In this section we examine our expressions in the limits in which the sine-Gordon theory maps onto a doubled Ising model ($\hat{\beta} = 1$) and the Gross-Neveu SU(2) model ($\hat{\beta} = \sqrt{2}$). We first examine the doubled Ising model.

5.1. Doubled Ising model

As demonstrated in Section 3, as $\hat{\beta}$ approaches 1, the perturbative expansion develops IR singularities. In our expressions, these singularities manifest themselves as poles in gamma functions. These poles are indicative of the logs that appear in the Ising spin-correlators in the scaling limit (see, for example, [3]). Specifically, an n th order pole term indicates an n th order log. The logs are obtained by expanding the powers of $|z|$ multiplying the pole terms. For example, writing $\hat{\beta}^2 = 1 + \epsilon$ with ϵ small, a typical situation will see the following expansion:

$$\frac{|z|^{1-\hat{\beta}^2}}{\hat{\beta}^2 - 1} = \frac{1}{\epsilon} - \log |z| + \dots \tag{5.1}$$

After this expansion is made, the divergent pieces (i.e. the $1/\epsilon$ term) are subtracted away. This can be understood in analogy with a cut-off method of regularisation. Throwing away the divergent terms amounts to no more than a specific choice of the cutoff. This procedure differs from that used by Dotsenko [6] where the powers of $|z|$ are not expanded and logs are substituted directly for the $1/\epsilon$ terms. This procedure is equivalent to ours in situations involving first-order logs. However it fails to produce the correct coefficients of higher-order logs that arise from overlapping divergences.

To test this regularisation technique, we examine a set of correlators simpler than the Ising correlator, free-fermion correlators. The free fermion action in Euclidean space-time reads

$$S = \frac{1}{4\pi} \int dx dt [\bar{\psi}_- \partial_z \psi_+ + \psi_- \partial_{\bar{z}} \psi_+ + im(\psi_- \bar{\psi}_+ - \bar{\psi}_- \psi_+)], \quad (5.2)$$

where we have introduced the Dirac spinors $\Psi_{\pm} = \begin{pmatrix} \bar{\psi}_{\pm} \\ \psi_{\pm} \end{pmatrix}$ which satisfy $\Psi_+ = \Psi_-^\dagger$ and employed an appropriate choice of gamma matrices². The correlators we are going to examine are

$$\langle \bar{\psi}_-(z, \bar{z}) \psi_+(0) \rangle = -2im K_0(mr), \quad (5.3)$$

$$\langle \psi_-(z, \bar{z}) \psi_+(0) \rangle = 2m \sqrt{\frac{\bar{z}}{z}} K_1(mr). \quad (5.4)$$

where the K's are the standard modified Bessel functions. These correlators can be derived from the following mode expansions:

$$\psi_+ / i\bar{\psi}_+ = \sqrt{m} \int_{-\infty}^{\infty} \frac{d\theta}{2\pi i} e^{\pm\theta/2} (c^\mp(\theta) e^{\mp ip(\theta) \cdot x} \mp d^\pm(\theta) e^{\pm ip(\theta) \cdot x}), \quad (5.5)$$

where the creation (+) and the destruction (−) operators obey the algebra

$$\{d^-(\theta), d^+(\theta')\} = \{c^-(\theta), c^+(\theta')\} = 4\pi^2 \delta(\theta - \theta').$$

As demonstrated by Mandelstam [10], the spinor components can be expressed in terms of the quasi-chiral components of the sine-Gordon field, ϕ^L and ϕ^R :

$$\begin{aligned} \psi_{\pm} &= \exp(\pm i\phi^L), \\ \bar{\psi}_{\pm} &= \exp(\mp i\phi^R). \end{aligned} \quad (5.6)$$

With these conventions we can make the following identification:

$$\langle e^{i\phi^R(\bar{z})} e^{i\phi^L(0)} \rangle = -2im K_0(mr). \quad (5.7)$$

At first order the l.h.s. of this equation reduces to

$$\begin{aligned} F(\hat{\beta} = 1) &\equiv \frac{\lambda}{2\pi} \int d^2w \langle e^{i\phi^R(\bar{z})} e^{i\phi^L(0)} e^{-i\Phi(w, \bar{w})} \rangle_{\text{CFT}} \\ &= -i \frac{\lambda}{2\pi} \int d^2w \frac{1}{(\bar{w} - \bar{z})} \frac{1}{w}. \end{aligned} \quad (5.8)$$

The i arises from taking apart $e^{i\Phi}$, i.e. $e^{\Phi(w, \bar{w})} = -ie^{-i\phi^L(w)} e^{-i\phi^R(\bar{w})}$.

This integral has a logarithmic divergence. To regulate it we continue the scaling dimensions of the operators (via continuing $\hat{\beta}$) as follows:

² $\gamma^0 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\gamma^1 = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$.

$$\begin{aligned}
 F(\hat{\beta}) &= \frac{\lambda}{2\pi} \int d^2w \langle e^{i\hat{\beta}\phi^R(\bar{z})} e^{i\hat{\beta}\phi^L(0)} e^{-i\hat{\beta}\Phi(w, \bar{w})} \rangle \\
 &= -i \frac{\lambda}{2\pi} \int d^2w \frac{1}{(\bar{w} - \bar{z})^{\hat{\beta}^2}} \frac{1}{w^{\hat{\beta}^2}}.
 \end{aligned} \tag{5.9}$$

Notice that the $U(1)$ charge of the vertex operators still sums to zero.

In general to extract the logarithmic divergence some caution needs to be exercised. If we expand in the above integral, $(\bar{w} - \bar{z})^{\hat{\beta}^2}$ via $(\bar{w} - \bar{z})^{1+\delta}$ and $w^{\hat{\beta}^2}$ via $w^{1+\epsilon}$, we expect the logarithmic divergence to manifest itself through a pole of the form $(\delta + \epsilon)^{-1}$ and not δ^{-1} or ϵ^{-1} . Identifying the necessary form of this pole term through a separate expansion of exponents will prove crucial to obtaining the correct answer in what follows.

Writing this integral in terms of Euclidean (x, t) coordinates, we easily evaluate it to be

$$F(1 + \delta, 1 + \epsilon) = 2i\lambda|z|^{-\delta-\epsilon}\Gamma(\delta + \epsilon)\Gamma(1 - \delta - \epsilon). \tag{5.10}$$

With this expansion of the exponents into ϵ and δ , λ 's scaling dimension becomes $1 - (\epsilon + \delta)/2$. So we may express λ as $\lambda = -m\mu^{-(\epsilon+\delta)/2}$ where μ is a mass scale fixed to be

$$\mu^{-(\epsilon+\delta)/2} = (1 - \frac{(\epsilon + \delta)}{2}\gamma + O(\epsilon\delta))m^{-(\epsilon+\delta)/2} \tag{5.11}$$

by Zamolodchikov's $\lambda - m$ relationship (see (2.2)). Taking $\epsilon, \delta \rightarrow 0$ in F then gives us

$$F(1 + \epsilon, 1 + \delta) = -im\mu^{(\delta+\epsilon)/2} \left(\frac{2}{\epsilon + \delta} - 2\log(m|z|) - 2\gamma \right). \tag{5.12}$$

As indicated before we drop the pole term. The above expression then reduces to

$$F(1) = 2im \left(\log\left(\frac{mr}{2}\right) + \gamma \right) \tag{5.13}$$

as $|z| = r/2$. In taking the limit $\epsilon, \delta \rightarrow 0$ we have not expanded out the $\mu^{(\epsilon+\delta)/2}$ term so as to preserve the scaling dimension of F . Expanding out $K_0(mr)$ to first order in m shows the two sides of (5.7) to be in agreement. We thus have gained some confidence that our methods of evaluating the perturbative terms are correct.

To illustrate the connection between our method of regularisation and a cut-off method of regularisation, consider regulating $F(1)$ with a cut-off $|w| < R$:

$$F(1) = -i \frac{\lambda}{2\pi} \int^R d^2w \frac{1}{(\bar{w} - \bar{z})^{\hat{\beta}^2}} \frac{1}{w^{\hat{\beta}^2}}. \tag{5.14}$$

Doing the integral we find

$$F(1) = 2i\lambda \log\left(\frac{R}{|z|}\right) + \text{const.} \tag{5.15}$$

Now R can be written as $R = abm^{-1}$ where a and b are dimensionless constants and a is to be taken to ∞ (as the cutoff is removed). So as $\lambda = -m$

$$F(1) = 2im \log(a^{-1}b^{-1}m|z|) + \text{const.} \quad (5.16)$$

To obtain a match between $F(1)$ and the known form of $\langle \bar{\psi}_-(z, \bar{z}) \psi_+(0) \rangle$, $\log(a)$ must be thrown away and a specific choice for b must be made. The throwing away of $\log(a)$ corresponds to our discarding $1/(\epsilon + \delta)$. However in our method, to our advantage, there is no need to fix the arbitrary constant, b .

The fermion correlators also provide an opportunity to test whether we are accurately taking into account the vacuum structure of the theory. The correlator $\langle \bar{\psi}_-(z, \bar{z}) \psi_+(0) \rangle$ has its first non-trivial contribution at second order in λ , the first order at which the bubbles contribute. Bosonizing this correlator and expanding to second order we obtain

$$\begin{aligned} 2m \sqrt{\frac{\bar{z}}{z}} K_1(mr) &= \langle \bar{\psi}_-(z, \bar{z}) \psi_+(0) \rangle \\ &= \langle e^{-i\phi^L(z, \bar{z})} e^{i\phi^L(0)} \rangle \\ &= \frac{1}{z} + \frac{1}{4\pi^2} K(1) + O(\lambda^4), \end{aligned} \quad (5.17)$$

where

$$K(1) = \frac{\lambda^2}{z} \int d^2w_1 d^2w_2 w_1 w_2^{-1} (w_1 - z)^{-1} (w_2 - z) |w_1 - w_2|^{-2}. \quad (5.18)$$

We note that the zeroth-order term of this correlator agrees with expansion of K_1 .

The integral K in (5.18) is divergent. (In fact it is quadratically divergent. If the bubbles were explicitly included we would only be facing a logarithmic divergence. That our evaluation of K still leads only to $\log$'s indicates we have correctly handled the bubbles.) Regulating $K(1)$ as before by continuing $\hat{\beta}$ we obtain

$$K(\hat{\beta}) = \frac{\lambda^2 |z|^{4-2\hat{\beta}^2}}{z^{\hat{\beta}^2}} \int d^2w_1 d^2w_2 w_1^{\hat{\beta}^2} w_2^{-\hat{\beta}^2} (w_1 - 1)^{-\hat{\beta}^2} (w_2 - 1)^{\hat{\beta}^2} |w_1 - w_2|^{-2\hat{\beta}^2}. \quad (5.19)$$

Again we must be careful in separately expanding the exponents. Doing so we have

$$\begin{aligned} K(\epsilon, \delta_1, \delta_2, \gamma_1, \gamma_2) &= \lambda^2 |z|^{2+\gamma_1-\gamma_2-\delta_1+\delta_2-2\epsilon} z^{-\hat{\beta}^2} \\ &\times \int d^2w_1 d^2w_2 w_1^{1+\gamma_1} w_2^{-1-\gamma_2} (w_1 - 1)^{-1-\delta_1} (w_2 - 1)^{1+\delta_2} |w_1 - w_2|^{-2-2\epsilon}. \end{aligned} \quad (5.20)$$

Now we expect the logarithmic divergence to appear as a pole in ϵ . Thus we should be able to set straightaway $\gamma_1 = \gamma_2 = \delta_1 = \delta_2 = 0$. However in the evaluation of the integral we will find that zeros of δ_1 and δ_2 will cancel poles in these same variables. So for now we set only γ_1 and γ_2 to 0. Then evaluating the integral using the techniques in Section 4, K equals

$$\begin{aligned}
 K(\epsilon, \delta_1, \delta_2) = & -4|z|^{2-\delta_1+\delta_2-2\epsilon} z^{-\hat{\beta}^2} \\
 & \times L[s(\delta_1)(s(\delta_2)+s(\epsilon-\delta_2))L_{21}+s(\delta_2)(s(\delta_1)-s(\epsilon+\delta_2))L_{12}], \\
 \end{aligned} \tag{5.21}$$

where L and L_{ij} equal

$$\begin{aligned}
 L = & \int_0^1 dw_1 \int_0^{w_1} dw_2 (w_1 - w_2)^{-1-\epsilon}, \\
 L_{ij} = & \int_1^\infty dw_i \int_1^{w_i} dw_j w_1 w_2^{-1} (w_1 - 1)^{-1-\delta_1} (w_2 - 1)^{1+\delta_2} (w_i - w_j)^{-1-\epsilon}. \tag{5.22}
 \end{aligned}$$

Performing the integrals, we find for K after taking $\delta_1, \delta_2 \rightarrow 0$:

$$\begin{aligned}
 K(\epsilon, \delta_1, \delta_2) = & -4\pi^2 |mz|^2 |\mu z|^{2-2\hat{\beta}^2} z^{-\hat{\beta}^2} \frac{1}{\epsilon} (1 + \epsilon) \\
 = & 4\pi^2 |mz|^2 z^{-\hat{\beta}^2} \left(-\frac{1}{\epsilon} + 2 \left(\log\left(\frac{mr}{2}\right) + \gamma - \frac{1}{2} \right) \right), \tag{5.23}
 \end{aligned}$$

where we have used $\mu^{-\epsilon} = (1 - \epsilon\gamma + O(\epsilon^2)) m^{-\epsilon}$ and $|z| = r/2$. Some of the details of this calculation may be found in Appendix C. We again preserve the correct scaling dimensions of $K(\hat{\beta})$ by not expanding $z^{-\hat{\beta}^2}$. Instead we expand only $|z|^{4-2\hat{\beta}^2}$, the z -dependence scaled out from the integral. It is this piece, coming from the integral, that we expect to generate the log's. Dropping the infinite piece in $K(1 + \epsilon)$, we find for $\langle \psi_-(z, \bar{z}) \psi_+(0) \rangle$

$$\begin{aligned}
 \langle \psi_-(z, \bar{z}) \psi_+(0) \rangle = & \frac{1}{z} + \frac{2}{z} |mz|^2 \left(\log\left(\frac{mr}{2}\right) + \gamma - \frac{1}{2} \right) + O(m^4) \\
 = & 2m \sqrt{\frac{\bar{z}}{z}} \left(\frac{1}{mr} + \frac{mr}{2} \left(\log\left(\frac{mr}{2}\right) + \gamma - \frac{1}{2} \right) \right) + O(m^4). \tag{5.24}
 \end{aligned}$$

Expanding out $K_1(mr)$, we find the two sides of (5.17) agree. Thus the bubble diagrams, even though not explicitly included, are taken into account.

Having demonstrated that our regularisation techniques work on fermion correlators, we now apply the same methods on the Ising spin correlator with some confidence. We recall the spin correlator is given by

$$\langle \sigma(z, \bar{z}) \sigma(0) \rangle^2 = \frac{1}{2} \left(G\left(\frac{1}{2}, -\frac{1}{2}\right) - G\left(\frac{1}{2}, \frac{1}{2}\right) \right). \tag{5.25}$$

From Wu et al. [3], the expansion in the scaling limit of the spin correlator is

$$\langle \sigma(z, \bar{z}) \sigma(0) \rangle^2 = \frac{1}{R^{1/2}} \left(1 + t\Omega + t^2 \left(\frac{\Omega^2}{4} + \frac{1}{8} \right) + \frac{t^3}{8} \Omega + \dots \right), \tag{5.26}$$

where $\Omega = \log(\frac{t}{8}) + \gamma$, $R \propto r$, and t , the scaling variable, is proportional to mr .

We can use the zeroth-order terms on both sides of (5.26) to fix R in terms of r . We find

$$R = 2r. \quad (5.27)$$

At the next order, (5.26) reduces to

$$\frac{t\Omega}{R^{1/2}} = \frac{1}{4\pi} L(1), \quad (5.28)$$

where $L(1)$ equals

$$\begin{aligned} L(1) &\equiv -\lambda \int d^2w \langle e^{i\Phi(z)/2} e^{i\Phi(0)} e^{-i\Phi(w)} \rangle \\ &= -\lambda \int d^2w |z|^{1/2} |w - z|^{-1} |w|^{-1}. \end{aligned} \quad (5.29)$$

As with the first-order term of the fermion correlator $\langle \bar{\psi}_-(z, \bar{z}) \psi_+(0) \rangle$, this integral has a logarithmic divergence. Regulating L as before by taking $L(1) \rightarrow L(\hat{\beta}) \rightarrow L(1 + \epsilon, 1 + \delta)$, we find

$$L(1 + \delta, 1 + \epsilon) = -\lambda |z|^{\hat{\beta}^2/2} |z|^{-\epsilon - \delta} \int d^2w |w - 1|^{-1 - \epsilon} |w|^{-1 - \delta}. \quad (5.30)$$

We expect the pole to appear as $1/(\delta + \epsilon)$. Doing the integral, as in Section 4, we see that it does:

$$L(\hat{\beta}) = 2\lambda |z|^{\hat{\beta}^2/2} |z|^{-\epsilon - \delta} s(1 + \tfrac{1}{2}\delta) B(\tfrac{1}{2} - \tfrac{1}{2}\delta, \tfrac{1}{2} - \tfrac{1}{2}\epsilon) B(1 - \tfrac{1}{2}\delta, \tfrac{1}{2}(\delta + \epsilon)). \quad (5.31)$$

Simplifying the above, we find

$$L(1 + \epsilon, 1 + \delta) = -2\pi m |z|^{\hat{\beta}^2/2} \mu^{\hat{\beta}^2 - 1} |\mu z|^{-\epsilon - \delta} \left(\frac{2}{\epsilon + \delta} + 4 \log(2) \right). \quad (5.32)$$

As before we do not expand $|z|^{\hat{\beta}^2/2} \mu^{\hat{\beta}^2 - 1}$ so as to preserve the correct scaling dimension. We then have

$$L(1) = 4\pi m |z|^{1/2} (\log(\frac{mr}{8}) + \gamma). \quad (5.33)$$

Taking the scaling variable, t , to equal mr we see $L(1) = \frac{4\pi t}{R^{1/2}} \Omega$. Comparing this with (5.26) we see we have agreement.

At second order, the perturbative expansion of the spin correlator gives

$$\frac{1}{R^{1/2}} t^2 \left(\frac{\Omega^2}{4} + \frac{1}{8} \right) = \frac{1}{8\pi^2} M(1) \quad (5.34)$$

where $M(1)$ equals

$$\begin{aligned}
M(1) &= \lambda^2 \int d^2w d^2y \langle e^{i\Phi(z)/2} e^{-i\Phi(0)/2} e^{i\Phi(w)} e^{i\Phi(y)} \rangle \\
&= \lambda^2 |z|^{3/2} \int d^2w d^2y |w-y|^{-2} |w-1|^{-1} |y-1|^{-1} |y|.
\end{aligned} \tag{5.35}$$

We will show (5.34) is valid to the first subleading log term.

$M(1)$ is (quadratically) divergent. Again, as with the second-order term of the fermion correlator, explicit inclusion of the bubble contribution leaves $M(1)$ logarithmically divergent. But, as before, explicit inclusion of the bubbles is unnecessary. To regulate, we continue $\hat{\beta}$. $M(\hat{\beta})$ is then

$$M(\hat{\beta}) = \lambda^2 |z|^{4-5\hat{\beta}^2/2} \int d^2w d^2y |w-y|^{-2\hat{\beta}^2} |w-1|^{\hat{\beta}^2} |w|^{-\hat{\beta}^2} |y-1|^{-\hat{\beta}^2} |y|^{\hat{\beta}^2}. \tag{5.36}$$

The logarithmic divergence of M arise solely from $|w-y|^{-2\hat{\beta}^2}$. We can thus set all other $\hat{\beta}$'s to 1 so obtaining

$$M(\hat{\beta}) = \lambda^2 |z|^{\hat{\beta}^2/2} |z|^{4-2\hat{\beta}^2} \int d^2w d^2y |w-y|^{-2\hat{\beta}^2} |w-1|^{-1} |y-1|^{-1} |y|. \tag{5.37}$$

We can do this integral as in Section 4 with the result

$$\begin{aligned}
M(\hat{\beta}) &= -4\lambda^2 |z|^{\hat{\beta}^2/2} |z|^{4-2\hat{\beta}^2} \left[J_+ \left(K_- + s(1/2 + \hat{\beta}^2) K_+ \right) \right. \\
&\quad \left. + J_- \left(K_+ + s(1/2 - \hat{\beta}^2) K_- \right) \right], \\
M(\hat{\beta}) &= -4\lambda^2 |z|^{\hat{\beta}^2/2} |z|^{4-2\hat{\beta}^2} \left[(J_+ - J_-) (K_- - K_+) \right. \\
&\quad \left. + O(\epsilon^2) \times K_- + O(\epsilon^2) \times K_+ \right],
\end{aligned} \tag{5.38}$$

where $\hat{\beta}^2 = 1 + \epsilon$ and

$$\begin{aligned}
K_{\pm}(\hat{\beta}) &= \pm \frac{1}{2} B(1 \pm 1/2, \hat{\beta}^2 - 1) \Gamma(1 - \hat{\beta}^2) \Gamma(\hat{\beta}^2 - 1) (\hat{\beta}^2 - 1) \\
&\quad \times {}_3F_2(1 \pm 1/2, \hat{\beta}^2, \hat{\beta}^2 - 1, \hat{\beta}^2 \pm 1/2, 2, 1),
\end{aligned} \tag{5.39}$$

$$\begin{aligned}
J_{\pm}(\hat{\beta}) &= B(1 \pm 1/2, 2 - \hat{\beta}^2) B(1 - \hat{\beta}^2, 1 \pm 1/2) \\
&\quad \times {}_3F_2(\pm 1/2, 2 - \hat{\beta}^2, 1 \pm 1/2, 3 - \hat{\beta}^2 \pm 1/2, 2 - \hat{\beta}^2 \pm 1/2, 1).
\end{aligned} \tag{5.40}$$

Setting $\hat{\beta}^2 = 1 + \epsilon$ everywhere we find

$$\begin{aligned}
K_- - K_+ &= \frac{1}{\epsilon^2} (1 + \epsilon 2 \log 2 + O(\epsilon^2)), \\
J_+ - J_- &= -\frac{\pi^2}{2} (1 + \epsilon 2 \log 2 + O(\epsilon^2)).
\end{aligned} \tag{5.41}$$

We note that the expression for $J_+ - J_-$ has only been established numerically (to 6 decimal places). Some of the details of this calculation may be found in Appendix D.

Putting everything together we find for $M(1 + \epsilon)$

$$\begin{aligned} M(1 + \epsilon) &= 4|z|^{4-5\hat{\beta}^2/2} \lambda^2 \frac{1}{\epsilon^2} \left(\frac{\pi^2}{2} + \epsilon 2\pi^2 \log 2 + O(\epsilon^2) \right) \\ &= \frac{2\pi^2}{\epsilon^2} |z|^{-\hat{\beta}^2/2} |zm|^2 (\epsilon^2 (2\Omega^2 + \text{const.}) + O(\epsilon)) \end{aligned} \quad (5.42)$$

where in the last line we have substituted $-m\mu^{1-\hat{\beta}^2}$ for λ . The presence of the $1/\epsilon^2$ term indicates, as expected, that the leading log term is $\log^2$. Substituting in t and $R^{1/2}$, subtracting the infinite terms, and dropping all but the $\log^2$ and log terms, we find

$$M(1) = \frac{t^2}{R^{1/2}} (2\pi^2 \Omega^2 + \text{const.}). \quad (5.43)$$

Comparing with (5.26) we find the coefficients of both the leading and subleading log's at second order agree.

The ability to reproduce the non-trivial behavior of the Ising spin correlators provides us with a degree of confidence that our expressions for the correlators away from $\hat{\beta} = 1$ are correct.

5.2. Gross-Neveu $SU(2)$

We now go on to demonstrate that our methods of evaluating the sine-Gordon correlators reproduce the known behaviour of the Gross-Neveu model (sine-Gordon at $\hat{\beta}^2 = 2$). Specifically we demonstrate that we are able to reproduce the known β -functions for λ and $\hat{\beta}$ governing the Kosterlitz-Thouless transition to lowest order, as say calculated by Amit et al. [11] and Boyanovsky [12].

As shown in Section 3, the sine-Gordon theory develops UV singularities as $\hat{\beta}^2 \rightarrow 2$. As with Ising at $\hat{\beta} = 1$, these singularities manifest themselves as poles in gamma functions. And similarly to Ising, we regulate the correlators by continuing away from $\hat{\beta}^2 = 2$ via $\hat{\beta}^2 = 2 - \epsilon$. However unlike Ising, we handle these singularities, because they are UV, with a conventional renormalisation.

In order to facilitate comparison with Amit et al. [11] and Boyanovsky [12] we employ a sine-Gordon action equivalent to theirs:

$$S = -\frac{1}{4\pi} \int d^2z \left[\partial_z \Phi_0 \partial_{\bar{z}} \Phi_0 + \frac{2\lambda_0}{\hat{\beta}_0^2} \cos(\hat{\beta}_0 \Phi_0) \right], \quad (5.44)$$

where the free propagator is

$$\langle \Phi_0(z) \Phi_0(0) \rangle = -\log(4z\bar{z}). \quad (5.45)$$

The difference in the propagators amounts to a difference in how its IR singularities are cured.

To renormalise the theory we employ the following renormalisation prescription:

$$\begin{aligned}
 \Phi_0^2 &= Z_\Phi \Phi_R^2; \\
 \hat{\beta}_0^2 &= Z_\Phi^{-1} \hat{\beta}_R^2; \\
 (\cos(\hat{\beta}\Phi))_0 &= \mu^{2-\hat{\beta}_0^2} Z_C (\cos(\hat{\beta}\Phi))_R; \\
 \lambda_0 &= \mu^{2-\hat{\beta}_0^2} Z_\lambda \lambda_R;
 \end{aligned} \tag{5.46}$$

where the bare quantities are listed on the left-hand side and the renormalized quantities on the right-hand side. λ_R is understood to be dimensionless. This prescription, like Amit et al.'s and Boyanovsky's, involves a trivial renormalisation of the $\hat{\beta}$ so as to ensure $\hat{\beta}_0 \Phi_0 = \hat{\beta}_R \Phi_R$. However it differs from their's in that it introduces a wavefunction renormalisation of the potential, $\cos(\hat{\beta}\Phi)$. That this renormalisation is necessary is evident from considering the correlator $\langle \cos(\hat{\beta}\Phi(z)) \cos(\hat{\beta}\Phi(0)) \rangle$. It possesses singularities at $O(\lambda^2)$, the only way which to remove is through a wavefunction renormalisation. A consequence of this is that Boyanovsky's [12] computation of the β -function for λ misses a contribution. (We note that Amit et al.'s [11] calculation of β_λ at higher orders was already known to be flawed because of his method of regulating the theory. See C. Lovelace [13] for a detailed discussion.)

The wavefunction renormalisation for Φ , Z_Φ , is found through calculating the correlator

$$\langle \partial_z \Phi_0(z) \partial_z \Phi_0(0) \rangle. \tag{5.47}$$

To compute this correlator, we employ a trick. We can write

$$\begin{aligned}
 \langle \partial_z \Phi_0(z) \partial_z \Phi_0(0) \rangle &= \lim_{\alpha \rightarrow 0} \frac{1}{\alpha^2} \partial_x \partial_y \langle e^{i\alpha \Phi_0(x)} e^{-i\alpha \Phi_0(y)} \rangle \Big|_{\substack{x=z \\ y=0}} \\
 &= \lim_{\alpha \rightarrow 0} \frac{1}{\alpha^2} \partial_x \partial_y \left[|x-y|^{-2\alpha^2} + \frac{\lambda_0^2}{16\pi^2 \gamma^2} \frac{r^{4-2\hat{\beta}_0^2-2\alpha^2}}{16} \right. \\
 &\quad \left. \times \int d^2 w_1 d^2 w_2 \frac{|w_1-1|^{2\gamma\alpha}}{|w_2-1|^{2\gamma\alpha}} \frac{|w_2|^{2\gamma\alpha}}{|w_1|^{2\gamma\alpha}} \frac{1}{|w_1-w_2|^{-2\hat{\beta}_0^2}} \right] \Big|_{\substack{x=z \\ y=0}},
 \end{aligned} \tag{5.48}$$

where $\gamma = \hat{\beta}_0$ (as will be seen shortly, there is good reason to introduce this extra notation) and $z = re^{i\phi}/2$. The above integral is one encountered in Subsection 4.2. So using previous results and taking the $\alpha \rightarrow 0$ limit we find

$$\langle \partial_z \Phi_0(z) \partial_z \Phi_0(0) \rangle = -\left(\frac{re^{i\phi}}{2}\right)^{-2} - \frac{\lambda_0^2}{64\pi^2 \gamma^4} (2-\hat{\beta}_0^2)(1-\hat{\beta}_0^2)(re^{i\phi})^{-\hat{\beta}_0^2} (re^{-i\phi})^{2-\hat{\beta}_0^2} L, \tag{5.49}$$

where L is

$$L = -\frac{8\pi^2 \gamma^2 s(\hat{\beta}_0^2)}{(1-\hat{\beta}_0^2)(2-\hat{\beta}_0^2)^2}. \tag{5.50}$$

To the order we are working, we may replace $\hat{\beta}_0$ with $\hat{\beta}_R$.

To exhibit the divergence in the $O(\lambda_0^2)$ term, one cannot simply write $\hat{\beta}_R^2 = 2 - \epsilon$ and expand. Our renormalization scheme involves a subtlety. We are using the distance ϵ from the critical point $\hat{\beta}_R^2 = 2$ both as a regularizer (L has a $1/\epsilon$ pole) and as a renormalization group parameter. We must be very careful not to mix these two uses together. How this might happen becomes most clear if we imagine that we are using a cut-off, Λ , to regularize the theory. Under this regulation scheme, the $O(\lambda_0^2)$ term of $\langle \partial_z \Phi_0(z) \partial_z \Phi_0(0) \rangle$ takes the form

$$A \log \Lambda + B\epsilon \log \Lambda + C + O(\epsilon^2 \log \Lambda). \quad (5.51)$$

Both $A \log \Lambda$ and $B\epsilon \log \Lambda$ will contribute to the β -functions. In our regularization scheme, $\log \Lambda$ is replaced with $1/\epsilon$. But this leads us, if we are naive, to confuse $B\epsilon \log \Lambda$ with the constant term C , and so get the β -functions wrong.

To avoid this, we must first identify the terms in the integrand that produce the divergence. In this case, the sole term is $|w_1 - w_2|^{-2\hat{\beta}_R^2}$. Expanding the exponent of this term alone (via $\hat{\beta}_R^2 = 2 - \epsilon$) leaves the $O(\lambda_0^2)$ term as

$$f(\hat{\beta}_R^2) \frac{1}{\epsilon} + C. \quad (5.52)$$

There is no confusion of $B\epsilon \log \Lambda$ with C : this confusion only arises if we allow the $1/\epsilon$ from $|w_1 - w_2|^{-2\hat{\beta}_R^2}$ to be canceled by the ϵ 's from the expansion of the other $\hat{\beta}_R^2$'s.

To facilitate the expansion of $\hat{\beta}_R^2$ from $|w_1 - w_2|^{-2\hat{\beta}_R^2}$ alone, we have written the correlator with γ replacing all occurrences of $\hat{\beta}_R^2$ other than $|w_1 - w_2|^{-2\hat{\beta}_R^2}$. Making this expansion, we find

$$\langle \partial_z \Phi_0(z) \partial_z \Phi_0(0) \rangle = -\left(\frac{re^{i\phi}}{2}\right)^{-2} + \frac{\lambda_R^2}{8\gamma^2\epsilon} (re^{i\phi})^{-2} + \text{finite term}. \quad (5.53)$$

We then find Z_Φ to be

$$Z_\Phi = 1 - \frac{\lambda_R^2}{32\gamma^2\epsilon} + O(\lambda^4) \quad (5.54)$$

where we have used minimal subtraction.

With this, we can calculate the β -function for β :

$$\beta_\beta = \mu \frac{\partial \hat{\beta}_R}{\partial \mu} = \frac{\lambda_R^2}{32\hat{\beta}_R} + O(\lambda_R^4). \quad (5.55)$$

Defining $\delta = -\epsilon/2$, we can recast the above in a form directly comparable with Amit et al.'s [11] and Boyanovsky's [12] result

$$\beta_\delta = \hat{\beta}_R \beta_{\hat{\beta}} = \frac{\lambda_R^2}{32} + O(\lambda^4). \quad (5.56)$$

The terms of these β -functions agree with their previous results. In particular, there are no terms of the form $O(\delta^n \lambda_R^2)$, $n > 0$.

We can compute Z_C through the correlator $\langle e^{i\beta\Phi} e^{-i\beta\Phi} \rangle$ as we expect by hermiticity for $e^{i\beta\Phi}$ to have the same wavefunction renormalization as $\cos \beta\Phi$. (Because its divergences are solely disconnected, the correlator $\langle e^{i\beta\Phi} e^{i\beta\Phi} \rangle$ does not contribute to the renormalization of $\langle \cos \beta\Phi \cos \beta\Phi \rangle$.)

Now $\langle e^{i\hat{\beta}_0\Phi_0(z)} e^{i\hat{\beta}_0\Phi_0(0)} \rangle$ equals

$$\langle e^{i\hat{\beta}_0\Phi_0(z)} e^{i\hat{\beta}_0\Phi_0(0)} \rangle = \mu^{2\hat{\beta}_0^2-4} \left[\mu^{4-2\hat{\beta}_0^2} r^{-2\hat{\beta}_0^2} + \frac{1}{256\pi^2} M \right], \quad (5.57)$$

where M is

$$M = \frac{\lambda_0^2 \mu^{4-2\hat{\beta}_0^2} r^{4-4\hat{\beta}_0^2}}{\hat{\beta}_0^4} \int d^2 w_1 d^2 w_2 \frac{|1-w_1|^{2\hat{\beta}_0^2} |w_2|^{2\hat{\beta}_0^2}}{|1-w_2|^{2\hat{\beta}_0^2} |w_1|^{2\hat{\beta}_0^2}} \frac{1}{|w_1-w_2|^{2\hat{\beta}_0^2}}. \quad (5.58)$$

To the order we are working, we may replace in M , λ_0 with $\mu^\epsilon \lambda_R$ and $\hat{\beta}_0$ with $\hat{\beta}_R$. Again we must only expand the exponent of the terms that lead to divergences, namely $|w_2-1|^{-2\hat{\beta}_0^2}$, $|w_1|^{-2\hat{\beta}_0^2}$, and $|w_1-w_2|^{-2\hat{\beta}_0^2}$. However in this case the divergence structure is more complicated. There are subdivergences that will lead to non-local divergences, and there are disconnected divergences (from $|w_2-1|^{-2\hat{\beta}_0^2}$ and $|w_1|^{-2\hat{\beta}_0^2}$) that need to be identified and discarded. Because of this, we expand the exponents of the three terms separately:

$$|w_2-1|^{-4+2\delta_2}, \quad |w_1|^{-4+2\delta_1}, \quad |w_1-w_2|^{-4+2\epsilon}. \quad (5.59)$$

As we are not interested in $O(\epsilon \lambda_R^3)$ corrections to β_λ , we can replace all other occurrences of $\hat{\beta}_R^2$ in M with 2. So like I_2 in Subsection 4.2, we evaluate M to be

$$\begin{aligned} M &= \lambda_R^2 r^{-4} (\mu r)^{2(\delta_1+\delta_2+\epsilon)} \int d^2 w_1 d^2 w_2 \frac{|1-w_1|^4 |w_2|^4}{|1-w_2|^{-4+2\delta_2} |w_1|^{-4+2\delta_1} |w_1-w_2|^{-4+2\epsilon}} \\ &= 4\lambda_R^2 r^{-4} (\mu r)^{2(\delta_1+\delta_2+\epsilon)} s(\delta_2) s(\epsilon) B(\delta_1+\epsilon, -1+\delta_2) \\ &\quad \times B(-1+\epsilon, -1+\delta_1) B(-(\delta_1+\delta_2+\epsilon), -1+\delta_2) B(1-\delta_1-\epsilon, -1+\epsilon) \\ &\quad \times {}_3F_2(-2, -1+\delta_1, \delta_1+\epsilon, -2+\delta_1+\epsilon, -1+\delta_1+\delta_2+\epsilon, 1) \\ &\quad \times {}_3F_2(-2, -(\delta_1+\delta_2+\epsilon), 1-\delta_1-\epsilon, -1-\delta_1-\epsilon, -\delta_1, 1). \end{aligned} \quad (5.60)$$

The exponent of the mass scale is adjusted so as to take into account the separate expansion of the exponents of the terms of the integrand. In this way the overall dimension of M is kept the same. Expanding M in δ_1 , δ_2 , and ϵ , we have

$$\begin{aligned} M &= 4\pi^2 \lambda_R^2 r^{-4} \left(\frac{1}{\delta_1 \delta_2} + \frac{1}{\delta_1 \epsilon} + \frac{1}{\delta_2 \epsilon} \right) \left(1 - \frac{\delta_1 \delta_2 + \delta_2 \epsilon + \delta_1 \epsilon}{\delta_1 + \delta_2 + \epsilon} \right) \\ &\quad + 16\pi^2 \lambda_R^2 r^{-4} \log \mu r \left(\frac{1}{\epsilon} + \frac{1}{\delta_1} + \frac{1}{\delta_2} \right) + \text{finite pieces}. \end{aligned} \quad (5.61)$$

We have thrown away terms of the form $\epsilon/\delta_1 \delta_2$ as these only make a contribution to β_λ at $O(\epsilon \lambda_R^3)$. Such terms show the importance of separate expansions of the exponents

producing the divergences so as to not confuse the divergences. Only in this way can terms like $\epsilon/\delta_1\delta_2$ not be mistaken for $1/\epsilon$. Throwing away the disconnected divergences, those involving poles of δ_1 and δ_2 alone, M becomes

$$M = \lambda_R^2 r^{-4} \pi^2 \left(4 \frac{-3}{\delta_1 + \delta_2 + \epsilon} + \frac{16}{\epsilon} \log \mu r \right). \quad (5.62)$$

We then take δ_1 and δ_2 to equal ϵ (permissible as we have now eliminated the disconnected divergences and there is no longer any need to distinguish divergences):

$$M = \lambda_R^2 r^{-4} \pi^2 \left(-\frac{4}{\epsilon} + \frac{16}{\epsilon} \log \mu r \right). \quad (5.63)$$

The non-local $\epsilon^{-1} \log \mu r$ term is a result of a subdivergence. If sine-Gordon at $\hat{\beta}_R^2 = 2$ is renormalizable, it must be canceled off. In fact it is by the zeroth-order term:

$$\begin{aligned} (\mu r)^{4-2\hat{\beta}_0^2} &= (\mu r)^{2(1-Z_\Phi^{-1})+\epsilon Z_\Phi^{-1}} \\ &= 1 - \frac{1}{16\epsilon} \lambda_R^2 \log \mu r + \text{finite pieces at } O(\lambda_R^2). \end{aligned} \quad (5.64)$$

So $\langle e^{i\hat{\beta}_0\Phi_0(z)} e^{i\hat{\beta}_0\Phi_0(0)} \rangle$ equals

$$\langle e^{i\hat{\beta}_0\Phi_0(z)} e^{i\hat{\beta}_0\Phi_0(0)} \rangle = r^{-4} + \frac{1}{64\epsilon} r^{-4} \lambda_R^2 + \text{finite pieces at } O(\lambda_R^2), \quad (5.65)$$

and Z_C is found to be

$$Z_C = 1 - \frac{1}{128\epsilon} \lambda_R^2. \quad (5.66)$$

Now knowing the wavefunction renormalization of $\cos(\hat{\beta}\Phi)$ and Φ allows the computation of the coupling constant renormalization, Z_λ , through $\langle \partial_z \Phi_0(z) e^{i\hat{\beta}_0\Phi_0(0)} \rangle$, the correlator. This correlator may be evaluated via

$$\langle \partial_z \Phi_0(z) e^{i\hat{\beta}_0\Phi_0(0)} \rangle = \lim_{\alpha \rightarrow 0} \frac{1}{i\alpha} \partial_z \langle e^{i\alpha\Phi_0(z)} e^{i(\hat{\beta}_0-\alpha)\Phi_0(0)} \rangle. \quad (5.67)$$

The correlator on the r.h.s. to $O(\lambda_0^3)$ equals

$$\langle e^{i\alpha\Phi_0(z)} e^{i(\hat{\beta}_0-\alpha)\Phi_0(0)} \rangle = -\mu^{\hat{\beta}_0^2-2} \left[\frac{1}{4\pi\hat{\beta}_0^2} I_1 + \frac{1}{128\pi^3\hat{\beta}_0^6} I_3 \right], \quad (5.68)$$

where

$$I_1 = \frac{1}{4} \mu^{2-\hat{\beta}_0^2} \lambda_0 r^{2-2\alpha^2-2\hat{\beta}_0^2} \int d^2 w |1-w|^{-2\alpha\hat{\beta}_0} |w|^{2\alpha\hat{\beta}_0-2\hat{\beta}_0^2}, \quad (5.69)$$

$$\begin{aligned} I_3 &= \frac{1}{64} \mu^{2-\hat{\beta}_0^2} \lambda_0^3 r^{6-4\hat{\beta}_0^2} \int d^2 w_1 d^2 w_2 d^2 w_3 |1-w_1|^{-2\alpha\hat{\beta}_0} |1-w_2|^{-2\alpha\hat{\beta}_0} |1-w_3|^{2\alpha\hat{\beta}_0} \\ &\quad \times |w_1|^{-2\hat{\beta}_0^2} |w_2|^{-2\hat{\beta}_0^2} |w_3|^{2\hat{\beta}_0^2} |w_1-w_2|^{2\hat{\beta}_0^2} |w_1-w_3|^{-2\hat{\beta}_0^2} |w_2-w_3|^{-2\hat{\beta}_0^2}. \end{aligned} \quad (5.70)$$

I_1 is easily evaluated. Using Subsection 4.1 we can write

$$I_1 = \mu^{2-\hat{\beta}_0^2} r^{2-2\alpha^2-2\hat{\beta}_0^2} \frac{\lambda_0}{2} \frac{s(\alpha\hat{\beta}_0)}{(\hat{\beta}_0^2-1)^2}. \quad (5.71)$$

With I_3 we have the same situation as with the evaluation of the correlator $\langle e^{i\beta\Phi} e^{-i\beta\Phi} \rangle$: there are both disconnected divergences and subdivergences. To keep track of these, we expand each term in the integrand leading to a divergence separately:

$$\begin{aligned} |w_1|^{-2\hat{\beta}_0^2} &= |w_1|^{-4+2\delta_1} \equiv |w_1|^{2c_1}, \\ |w_2|^{-2\hat{\beta}_0^2} &= |w_2|^{-4+2\delta_2} \equiv |w_2|^{2c_2}, \\ |w_1 - w_3|^{-2\hat{\beta}_0^2} &= |w_1 - w_3|^{-4+2\epsilon_1} \equiv |w_1 - w_3|^{2b_{13}}, \\ |w_2 - w_3|^{-2\hat{\beta}_0^2} &= |w_2 - w_3|^{-4+2\epsilon_2} \equiv |w_2 - w_3|^{2b_{23}}. \end{aligned} \quad (5.72)$$

We set all the remaining $\hat{\beta}_0^2$ in I_3 to 2. Doing so, I_3 can be evaluated in the $\alpha \rightarrow 0$ limit to be

$$\begin{aligned} I_3 &= \frac{1}{8} r^{-2} (r\mu)^{2(\epsilon_1+\epsilon_2+\delta_1+\delta_2)} \lambda_R^3 s(\epsilon_1) s(\epsilon_2) \\ &\quad \times [s(\alpha\hat{\beta}_0) (R(1,2,3) + R(2,1,3)) (S(3,2,1) + S(3,1,2)) \\ &\quad - s(\alpha\hat{\beta}_0) (R(2,3,1)S(1,3,2) + R(1,3,2)S(2,3,1))] \end{aligned} \quad (5.73)$$

where

$$\begin{aligned} R(i_1, i_2, i_3) &= B(1, 1+r_1) B(1+b_{i_2 i_3}, 1+r_2) B(1+b_{i_1 i_2}, 1+r_3) \\ &\quad \times {}_3F_2(-b_{i_1 i_3}, 1+r_2, 1+r_3, 2+r_2+b_{i_2 i_3}, 2+b_{i_1 i_2}+r_3, 1), \\ r_1 &= -4 - c_{i_1} - c_{i_2} - c_{i_3} - b_{i_1 i_3} - b_{i_2 i_3} - b_{i_1 i_2}, \\ r_2 &= -3 - c_{i_1} - c_{i_2} - b_{i_1 i_3} - b_{i_2 i_3} - b_{i_1 i_2}, \\ r_3 &= -2 - c_{i_1} - b_{i_1 i_2} - b_{i_1 i_3} \end{aligned} \quad (5.74)$$

and

$$\begin{aligned} S(i_1, i_2, i_3) &= B(1, 1+s_1) B(1+b_{i_1 i_2}, 1+s_2) B(1+b_{i_2 i_3}, 1+s_3) \\ &\quad \times {}_3F_2(-b_{i_1 i_3}, 1+s_2, 1+s_3, 2+s_2+b_{i_1 i_2}, 2+b_{i_2 i_3}+s_3, 1), \\ s_1 &= 2 + c_{i_1} + c_{i_2} + c_{i_3} + b_{i_1 i_3} + b_{i_2 i_3} + b_{i_1 i_2}, \\ s_2 &= 1 + c_{i_2} + c_{i_3} + b_{i_2 i_3}, \\ s_3 &= c_{i_3}. \end{aligned} \quad (5.75)$$

$c_1, c_2, b_{23} = b_{32}, b_{13} = b_{31}$ are defined in (5.72). The other variables are defined by $b_{12} = b_{21} = c_3 = 2$.

Expanding the expression in terms of δ_1 , δ_2 , ϵ_1 , and ϵ_2 we have

$$I_3 = \frac{1}{2} \pi^2 s (\alpha \hat{\beta}_0) r^{-2} (\mu r)^{2(\epsilon_1 + \epsilon_2 + \delta_1 + \delta_2)} \left[\frac{1}{\epsilon_1 \epsilon_2} + \frac{1}{\epsilon_2 \delta_2} + \frac{1}{\epsilon_1 \delta_1} + 2 \left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} \right) \right]. \quad (5.76)$$

Again, we have dropped terms such as $1/\delta_1 \delta_2$ that only lead to disconnected divergences and terms of the form ϵ_1/ϵ_2^2 and δ_1/ϵ_1^2 etc..

We can now put everything together and write down for $\langle \partial_z \Phi_0(z) e^{i\hat{\beta}_0 \Phi_0(0)} \rangle$ the following expression:

$$\langle \partial_z \Phi_0(z) e^{i\hat{\beta}_0 \Phi_0(0)} \rangle = -\frac{i}{8\hat{\beta}_R} \frac{r^{-2}}{z} \lambda_R Z_\lambda Z_\Phi^{1/2} \left[1 - \frac{5\lambda_R^2}{128\epsilon} \right]. \quad (5.77)$$

Again the non-local divergences arising from the renormalization of $\hat{\beta}_0$ in I_1 are canceled by those arising in I_3 from the double poles with the expansion of $(\mu r)^{2(\epsilon_1 + \epsilon_2 + \delta_1 + \delta_2)}$. Because $\langle \partial_z \Phi_R^2(z) (e^{i\hat{\beta}_R \Phi_R^2(0)})_R \rangle = Z_\Phi^{-1/2} Z_C^{-1} \langle \partial_z \Phi_0(z) e^{i\hat{\beta}_0 \Phi_0(0)} \rangle$ is finite, we see that Z_λ equals

$$Z_\lambda = Z_C \left(1 + \frac{5\lambda_R^2}{128\epsilon} \right) \quad (5.78)$$

$$= 1 + \frac{\lambda_R^2}{32\epsilon}. \quad (5.79)$$

So β_λ equals

$$\beta_\lambda = \mu \frac{\partial \lambda_R}{\partial \mu} = -\epsilon \lambda_R + \frac{\lambda_R^3}{16} + O(\epsilon \lambda_R^3) + O(\lambda_R^5) \quad (5.80)$$

$$= 2\delta \lambda_R + \frac{\lambda_R^3}{16}. \quad (5.81)$$

The $O(\lambda_R^3)$ term calculated previously by Boyanovsky [12] was $\frac{\lambda_R^3}{32}$. The difference arises from the renormalization of the cosine potential.

6. Conclusion

We have shown how the short-distance expansion of some sine-Gordon correlation functions is well defined and reasonably tractable. The ultimate goal is to find some non-perturbative characterization of these correlation functions. We make two remarks. First of all, unlike the large-distance expansion, the short-distance expansions at and away from the free fermion point are of comparable complexity. Secondly, we have in no way utilized the integrability structure of the sine-Gordon theory, i.e. factorizable S -matrix, quantum inverse scattering method, etc. in developing the short-distance expansion. It would be very interesting to understand the consequences of integrability in this context.

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Appendix A. Evaluation of $K(i, j)$

In this appendix we evaluate the integrals $K(i, j)$ introduced in Subsection 4.2. Using (4.11) we can write $K(i, j)$ as

$$\begin{aligned} K(i, j) &= \int_0^1 dw_i w_i^{-2+\hat{\beta}^2} (1-w_i)^{\gamma_i} \int_0^{w_i} dw_j w_j^{-2+\hat{\beta}^2} (1-w_j)^{\gamma_j} (w_i-w_j)^{-\hat{\beta}^2} \\ &= B(\gamma_i+1, \hat{\beta}^2-2) B(1-\hat{\beta}^2, \hat{\beta}^2-1) \\ &\quad \times \sum_{k=0}^{\infty} \frac{(-\gamma_j)_k (\hat{\beta}^2-2)_k (\hat{\beta}^2-1)_k}{k! (\gamma_i+\hat{\beta}^2-1)_k (0)_k} \end{aligned} \quad (\text{A.1})$$

where $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$, $(a)_k = \Gamma(a+k)/\Gamma(a)$, and the sum $\sum_k$ is the standard hypergeometric function,

$${}_3F_2(-\gamma_j, \hat{\beta}^2-2, \hat{\beta}^2-1, \gamma_i+\hat{\beta}^2-1, 0, 1). \quad (\text{A.2})$$

However this form is somewhat problematic as a pole in ${}_3F_2$ is multiplying a zero in a β -function, B. To make sense of this expression we continue $w_j^{-2+\hat{\beta}^2}$ to $w_j^{-2+\hat{\beta}^2+\epsilon}$ in the above integrand and take the limit $\epsilon \rightarrow 0$. With this continuation, $K(i, j)$ equals

$$\begin{aligned} K(i, j) &= B(\gamma_i+1, \hat{\beta}^2-2) B(1-\hat{\beta}^2, \hat{\beta}^2-1+\epsilon) \\ &\quad \times {}_3F_2(-\gamma_j, \hat{\beta}^2-2, \hat{\beta}^2-1, \gamma_i+\hat{\beta}^2-1, \epsilon, 1). \end{aligned} \quad (\text{A.3})$$

Using the identity

$$\begin{aligned} \lim_{\zeta \rightarrow -n} \frac{{}_3F_2(\alpha, \beta, \gamma, \delta, \zeta, z)}{\Gamma(\zeta)} \\ &= \frac{\alpha(\alpha+1) \cdots (\alpha+n) \beta(\beta+1) \cdots (\beta+n) \gamma(\gamma+1) \cdots (\gamma+n)}{(n+1)! \delta(\delta+1) \cdots (\delta+n)} \\ &\quad \times z^{n+1} {}_3F_2(\alpha+n+1, \beta+n+1, \gamma+n+1, \delta+n+1, 2, z), \end{aligned} \quad (\text{A.4})$$

and then taking the limit $\epsilon \rightarrow 0$, we obtain

$$\begin{aligned} K(i, j) &= B(1+\gamma_i, \hat{\beta}^2-1) \Gamma(1-\hat{\beta}^2) \Gamma(\hat{\beta}^2-1) (-\gamma_j) (\hat{\beta}^2-1) \\ &\quad \times {}_3F_2(1-\gamma_j, \hat{\beta}^2-1, \hat{\beta}^2, \hat{\beta}^2+\gamma_i, 2, 1), \end{aligned} \quad (\text{A.5})$$

where we have also used the relation $x\Gamma(x) = \Gamma(x+1)$.

Appendix B. Analytic continuation of ${}_3F_2$

The Gaussian hypergeometric function ${}_3F_2(\alpha, \beta, \gamma, \delta, \zeta, 1)$ is, in general, a meromorphic function of the variables $\alpha, \beta, \gamma, \delta$, and ζ . However its sum representation (as say given in Appendix A) is not necessarily convergent for all values of these variables. The convergence of the sum is determined by defining the “excess” parameter, s , by

$$s = \delta + \zeta - (\alpha + \beta + \gamma). \quad (\text{B.1})$$

If $s > 0$ the sum converges. However we will often encounter situations where $s < 0$ and the sum representation is no good. To get around this problem analytic continuation formulae are needed. The only one that will prove to be necessary is an identity known as Dixon’s theorem:

$${}_3F_2(\alpha, \beta, \gamma, \delta, \zeta, 1) = \frac{\Gamma(\delta)\Gamma(\zeta)\Gamma(s)}{\Gamma(\alpha)\Gamma(s+\beta)\Gamma(s+\gamma)} {}_3F_2(\delta-\alpha, \zeta-\alpha, s, s+\beta, s+\gamma, 1). \quad (\text{B.2})$$

The convergence of the hypergeometric function of the right-hand side is then determined by the positivity of α . Notice that because ${}_3F_2$ is invariant under permutations of α, β , and γ and permutations of δ and ζ , variations (144 in total – not all necessarily distinct) of this identity exist.

Appendix C. Taking $\epsilon \rightarrow 1$ in $\langle \psi_-(z, \bar{z}) \psi_+(0) \rangle$

In taking this limit in $\langle \psi_-(z, \bar{z}) \psi_+(0) \rangle$, the only subtlety comes in calculating the L_{ij} . From (5.22) we have

$$L_{ij} = \int_1^\infty dw_i \int_1^{w_i} dw_j w_1 w_2^{-1} (w_1 - 1)^{-1-\delta_1} (w_2 - 1)^{1+\delta_2} (w_i - w_j)^{-1-\epsilon}. \quad (\text{C.1})$$

This integral is itself straightforward to evaluate:

$$\begin{aligned} L_{ij} = & \text{B}(-1 - \delta_2 + \delta_1 + \epsilon, 1 + (-1)^j(1 + \delta_j)) \text{B}(-\epsilon, \epsilon + (-1)^j \delta_i) \\ & \times {}_3F_2((-1)^j(1 + \delta_i), -1 + \delta_1 - \delta_2 + \epsilon, \epsilon + (-1)^j \delta_i, \\ & \delta_1 - \delta_2 + \epsilon + (-1)^j(1 + \delta_j), -(-1)^j \delta_i, 1). \end{aligned} \quad (\text{C.2})$$

Now in making its contribution to $\langle \psi_-(z, \bar{z}) \psi_+(0) \rangle$, L_{ij} is multiplied by $s(\delta_j)$. But L_{12} has no poles in δ_2 and so as $\delta_2 \rightarrow 0$, its contribution vanishes. We are then left with the evaluation of L_{21} :

$$\begin{aligned} L_{21} = & \text{B}(-1 - \delta_2 + \delta_1 + \epsilon, -\delta_1) \text{B}(-\epsilon, \epsilon - \delta_2) \\ & \times {}_3F_2(-1 - \delta_2, -1 + \delta_1 - \delta_2 + \epsilon, \epsilon - \delta_2, -1 - \delta_2 + \epsilon, -\delta_2, 1) \end{aligned}$$

$$= \Gamma(-\epsilon)\Gamma(-\delta_1) \sum_{k=0}^{\infty} \frac{(-1-\delta_2)_k \Gamma(k+\delta_1-\delta_2-1+\epsilon) \Gamma(k+\epsilon-\delta_2)}{k! \Gamma(k-\delta_2) \Gamma(k-1-\delta_2+\epsilon)}. \quad (\text{C.3})$$

Now taking δ_1 and δ_2 to 0 (the log divergence is solely encoded in ϵ) all but the $k = 1$ term vanish leaving us with

$$L_{21} = -\Gamma(-\epsilon)\Gamma(-\delta_1)\Gamma(1+\epsilon) = -\frac{1}{\epsilon\delta_1}(1 + O(\epsilon^2)). \quad (\text{C.4})$$

The remaining pole in δ_1 is canceled off by the $s(\delta_1)$ appearing in (5.21), leaving alone a pole in ϵ .

Appendix D. Taking $\epsilon \rightarrow 0$ in $\langle \sigma(z, \bar{z}) \sigma(0) \rangle^2$

We begin by evaluating the $K_{\pm}$. As in Appendix C we focus on the hypergeometric functions that appear in these two expressions. We have the following function:

$$S_{K_{\pm}}(\hat{\beta}) = {}_3F_2(1 \pm 1/2, \hat{\beta}^2, \hat{\beta}^2 - 1, \hat{\beta}^2 \pm 1/2, 2, 1). \quad (\text{D.1})$$

To $O(\epsilon)$, $S_{K_{\pm}}(\hat{\beta})$ is

$$\begin{aligned} S_{K_{\pm}}(\hat{\beta}) &= {}_3F_2(1 \pm 1/2, 1, \epsilon, 1 \pm 1/2, 2, 1) + O(\epsilon^2) \\ &= \sum_{k=0}^{\infty} \frac{(\epsilon)_k (1)_k}{(2)_k k!} + O(\epsilon^2) \\ &= \frac{\Gamma(2)\Gamma(1-\epsilon)}{\Gamma(2-\epsilon)\Gamma(1)} + O(\epsilon^2) \\ &= 1 + \epsilon + O(\epsilon^2), \end{aligned} \quad (\text{D.2})$$

where we have used the Gaussian summation formula

$${}_2F_1(a, b, c, 1) = \sum_{k=0}^{\infty} \frac{(a)_k (b)_k}{(c)_k k!} = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}. \quad (\text{D.3})$$

We now move on to the more difficult evaluation of $J_{\pm}$. Rather than evaluate each separately we restrict ourselves to evaluating their difference $J_+ - J_-$. We can show straightforwardly that

$$J_+ - J_- = 2 \frac{\Gamma^2(1-\epsilon)\Gamma(-\epsilon)\Gamma(3/2)}{\Gamma(2-2\epsilon)\Gamma(3/2-\epsilon)} [\epsilon L_1 - L_2] + O(\epsilon^2), \quad (\text{D.4})$$

where L_1 and L_2 are

$$L_1 = \sum_{k=0}^{\infty} \frac{(-\epsilon)_k (1)_k^2}{k! (2)_k (3/2)_k},$$

$$L_2 = \sum_{k=0}^{\infty} \frac{k(-\epsilon)_k (1-\epsilon)_k^2}{k! (2-2\epsilon)_k (3/2-\epsilon)_k}.$$

To evaluate L_1 we write

$$\begin{aligned} L_1 &= 1 - \epsilon \sum_{k=1}^{\infty} \frac{(1)_{k-1} (1)_k}{(3/2)_k (2)_k} \\ &= 1 - \epsilon \lim_{\delta \rightarrow 0} \sum_{k=2}^{\infty} \frac{(1+\delta)_{k-2} (1+\delta)_{k-1}}{(3/2)_{k-1} (2)_{k-1}} \\ &= 1 - \frac{\epsilon}{2} \lim_{\delta \rightarrow 0} \frac{\Gamma(\delta-1)\Gamma(\delta)}{\Gamma^2(1+\delta)} \sum_{k=2}^{\infty} \frac{(\delta-1)_k (\delta)_k}{(1/2)_k k!}. \end{aligned} \quad (\text{D.5})$$

Now using the Gaussian summation formula we find

$$L_1 = 1 - \frac{\epsilon}{2} \lim_{\delta \rightarrow 0} \frac{\Gamma(\delta-1)\Gamma(\delta)}{\Gamma^2(1+\delta)} \left[\frac{\Gamma(3/2-2\delta)\Gamma(1/2)}{\Gamma(3/2-\delta)\Gamma(1/2-\delta)} - 1 - 2\delta(\delta-1) \right]. \quad (\text{D.6})$$

Evaluating this limit we obtain

$$L_1 = 1 + \epsilon \left(\frac{\pi^2}{4} - 3 \right) + O(\epsilon^2). \quad (\text{D.7})$$

To evaluate L_2 we write it as

$$\begin{aligned} L_2 &= -\epsilon \sum_{k=1}^{\infty} \frac{k(1-\epsilon)_{k-1} (1-\epsilon)_k^2}{k! (2)_k (3/2-\epsilon)_k} - 2\epsilon^2 \sum_{k=1}^{\infty} \frac{k(1)_{k-1} (1)_k^2}{k! (2)_k (3/2)_k} (\psi(2+k) - \psi(2)) \\ &= -\epsilon L_{21} - 2\epsilon^2 L_{22}. \end{aligned} \quad (\text{D.8})$$

L_{21} can be rewritten as

$$\begin{aligned} L_{21} &= \sum_{k=0}^{\infty} \frac{(k+1)(1-\epsilon)_k (1-2\epsilon)_{k+1}}{(2)_{k+1} (3/2-\epsilon)_{k+1}} \\ &= \frac{(1-2\epsilon)}{4\epsilon^2} \sum_{k=2}^{\infty} \frac{(-\epsilon)_k (-2\epsilon)_k}{k! (1/2-\epsilon)_k} + \epsilon \sum_{k=2}^{\infty} \frac{(1)_{k-2} (1)_{k-1}}{(2)_{k-1} (3/2)_{k-1}} \\ &= L_{211} + L_{212}. \end{aligned} \quad (\text{D.9})$$

The first term, L_{211} , can be evaluated using the Gaussian summation formula to be

$$L_{211} = -1 + \frac{\pi^2}{4} - \frac{\epsilon}{4} (2\pi^2 - \psi''(1/2)). \quad (\text{D.10})$$

L_{212} can be evaluated in the same fashion as L_1 with the result

$$L_{212} = \epsilon \left(3 - \frac{\pi^2}{4} \right). \quad (\text{D.11})$$

We were only able to evaluate L_{22} numerically:

$$L_{22} = 3.055492\dots \approx \frac{\pi^2}{2} \log 2 - \frac{\pi^2}{4} - \frac{\psi''(1/2)}{8}. \quad (\text{D.12})$$

This approximation has been verified to six decimal places. Putting this altogether we find for L_2

$$L_2 = -\epsilon \left(\frac{\pi^2}{4} - 1 + \epsilon \left(3 - \frac{5}{4}\pi^2 + \pi^2 \log 2 \right) \right). \quad (\text{D.13})$$

So $J_+ - J_-$ equals

$$J_+ - J_- = -\frac{\pi^2}{2} (1 + 2\epsilon \log 2). \quad (\text{D.14})$$

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